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# INFERENCE AS CONSOLIDATION: REPLAY IN THE SILENT DEGREES OF FREEDOM

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## ABSTRACT

Brains consolidate memories not only in sleep but also, as recent causal evidence shows, through *local sleep*: brief, use-dependent off-periods of individual cortical circuits during wakefulness. Replay-based continual learning in artificial networks, by contrast, almost always consolidates in a dedicated offline phase or by interleaving replayed samples with the input stream. We ask whether a network trained by local, biologically constrained learning rules can consolidate *during inference*, with no offline phase at all. The construction has three parts. An *isolation* rule confines replay updates to hidden synapses that are invisible to the current input under  $k$ -winner-take-all ( $k$ -WTA) dynamics, with optimiser state advanced only inside the mask. A *refractory rotation* rule makes units that have just fired sit out the next competition, which widens the set of consolidable synapses. Two internal signals, a unit-level homeostatic pressure and a relative-novelty gate, decide when replay bursts fire and when rotation runs. This inverts the usual direction of non-interfering continual learning: rather than protecting past tasks while learning the present one, the hidden computation on the current input is held invariant (exactly for silent and suppressed units when the mask matches the weights being updated, and for all but 0.3% of waking samples per update in the implemented form) while past memories are written into the degrees of freedom the current batch leaves unused. On class-incremental split-MNIST the system reaches  $91.6 \pm 0.3\%$  with no offline phase, at or above the best offline-night schedule on two held-out splits and consistently above backpropagation with experience replay (BP+ER) and unmasked local replay, at about twice the night’s replay samples; in a single pass over the stream the lead widens (91.8% against 88.7% for BP+ER and 75.1% for the night). Rotation carries most of the gain; isolation adds the invariance guarantee and, at two-sample replay batches, avoids the severe failures that unmasked replay shows in half the seeds. The ordering transfers to split CIFAR-10, where a substrate decomposition attributes the remaining gap to BP+ER to dense activation rather than to credit assignment. Finally, a drive-referenced controller for the rule’s synaptic decay replaces per-dataset tuning with a single ratio target and, with locally learned skip synapses, holds a five-hidden-layer network at the two-layer level. We prove exactness for the isolation rule and release all code, checkpoints and diagnostics.

**Keywords** continual learning · replay · local sleep · biologically constrained learning ·  $k$ -winner-take-all

## 1 Problem Formulation

**Setting.** Let a stream of labelled examples  $(x_t, y_t) \in \mathbb{R}^d \times \{1, \dots, C\}$  be presented in  $T$  contiguous tasks  $\mathcal{T}_1, \dots, \mathcal{T}_T$ , where task  $\mathcal{T}_\tau$  contains only classes  $\mathcal{C}_\tau \subset \{1, \dots, C\}$ , the  $\mathcal{C}_\tau$  are disjoint, and no task identity is available at test time (class-incremental learning). A network  $f_\theta$  with a single shared  $C$ -way readout is trained on the stream and evaluated on all  $C$  classes after the final task,

$$A_{\text{seq}}(\theta) = \Pr_{(x,y) \sim \mathcal{D}_{\text{test}}} [\arg \max_c f_\theta(x)_c = y], \quad F = \frac{1}{T-1} \sum_{\tau=1}^{T-1} (a_{\tau,\tau} - a_{T,\tau}), \quad (1)$$

where  $a_{t,\tau}$  is accuracy on task  $\tau$  after learning task  $t$  and  $F$  is forgetting.

**Constraints.** We restrict the learner to a set  $\mathcal{B}$  of cortical constraints: no weight transport (feedback synapses are learned, never copied), errors carried by a bounded burst-like channel, Dale’s law, sparse population codes ( $k$ -WTA on wide layers), sparse distance-dependent connectivity, and a single forward–feedback sweep per example (no iterative relaxation). An episodic memory (“hippocampus”) of at most  $K$  raw examples with reservoir writing (Vitter, 1985) is permitted;  $K$  is a budget, not a free parameter.

**Objective.** Rather than a single accuracy number, we optimise a four-way trade-off

$$\max_{\theta} (A_{\text{seq}}, A_{\text{iid}}, -R, -E) \quad \text{subject to } \theta \in \mathcal{B}, \quad (2)$$

where  $A_{\text{iid}}$  is accuracy of the identical learner on the i.i.d. shuffle of the same data (the *static axis*: a brain-like mechanism must not damage ordinary learning),  $R$  is the replay cost in samples (synaptic work is reported separately), and  $E$  an arithmetic-energy proxy (Section 3). The central question of this paper is whether the offline consolidation phase can be removed entirely, so that consolidation occurs only *during* inference on the wake stream, without losing  $A_{\text{seq}}$ , and at what cost in  $A_{\text{iid}}$  and  $R$ .

## 2 Related Work

**Sleep-inspired consolidation in ANNs.** The Bazhenov line implements sleep as a global offline phase: networks are converted to spiking dynamics and driven by noise under local Hebbian plasticity, recovering old tasks after each new one (Tadros et al., 2022), after several tasks at once (Bazhenov et al., 2026), in spiking networks with coarse alternation of training and sleep (Golden et al., 2022), and in equilibrium-propagation networks combined with small rehearsal buffers (Kubo et al., 2025). Wake–sleep frameworks with NREM and REM stages (Sorrenti et al., 2025; Robinson et al., 2022) and engineering systems such as SIESTA (Harun et al., 2023) likewise schedule consolidation offline, and SESLR (Lin et al., 2026) appends a noise-enhanced sleep phase, in which a spiking classifier on a frozen extractor trains only on its replay buffer, to correct the recency bias left by online learning. In all of these, sleep is global, offline, and clocked. None confines consolidation to currently unused units during ongoing inference, and the bias correction such a phase delivers is what our plastic readout receives continuously from isolated replay.

**Gating, subnetwork and history-dependent competition methods.** Context-dependent gating (XdG) activates a fixed random subnetwork per task but requires task identity at train and test time (Masse et al., 2018); learned gates (Tilley et al., 2023), dendritic context vectors from input prototypes (Iyer et al., 2022), task-free sparsification by feedforward and top-down errors (Lässig et al., 2023), and gradient-recovered functional masks (McKee et al., 2026) remove the explicit label but derive the gate from the *input identity*. Use-history-dependent suppression of competition, in itself, has classical precedents: refractory winner-take-all dynamics in physiological models of competitive learning (Kaski and Kohonen, 1994), explicit refractory competitive learning in which a recent winner is temporarily barred from winning (Maeda and Miyajima, 1999), and, recently, spiking continual learners whose firing thresholds rise with each unit’s activation history (Shen et al., 2024) or whose top- $k$  selection is randomised (Shen et al., 2026) to diversify recruitment across tasks, the wake-side counterpart of our random-alternation control. Our rotation differs in *purpose*, not in *primitive*: recently wake-active units are benched for one competition specifically to enlarge a dynamically reallocated *replay channel*. Rotation creates plastic degrees of freedom in which consolidation can proceed concurrently with inference, rather than allocating representational resources across tasks.

**Null-space and parameter-isolation methods.** A mature line of work constructs non-interfering updates by protecting *previous* tasks: projecting new-task gradients orthogonally to old-task gradient or activation subspaces (Farajtabar et al., 2020; Saha et al., 2021), training in the null space of accumulated feature covariance (Wang et al., 2021), combining  $k$ -winner sparsity and heterogeneous dropout with such projections (Abbasi et al., 2022), or freezing pruned-and-frozen subnetworks (Golkar et al., 2019; Mallya and Lazebnik, 2018). Our construction inverts the protected direction: the *ongoing wake computation* is held invariant while *old-memory replay* is written into degrees of freedom that the current batch’s sparse support exposes as instantaneously invisible. No bases, SVDs, or task masks are stored, and the “null space” changes with every batch at no cost. McKee et al. (2026) note that optimiser state and decoupled weight decay can break structural guarantees even at zero gradient, and Bricken et al. (2023) analyse stale momentum in sparse learners; we build the corresponding requirement, mask-confined optimiser state, into the exactness construction. Replay *scheduling* has also been studied as a learned policy (Klasson et al., 2023); our contribution there is a specific unit-level use-pressure trigger and its empirical dissociation from novelty triggering.

**Complementary learning systems.** CLS-ER maintains fast/slow EMA copies of the weights as semantic memories (Arani et al., 2022) and DualNet trains a slow self-supervised learner beside a fast supervised one (Pham et al., 2021); both raise the value of each replayed sample but still replay at every step. Our contribution is orthogonal: we change *where* (isolated synapses), *when* (homeostatic bursts) and at what granularity (micro-batches) replay is applied.

**Pattern separation.** Cerebellum-, mushroom-body- and dentate-gyrus-style sparse expansions (Cayco-Gajic et al., 2017; Cayco-Gajic and Silver, 2019; Litwin-Kumar et al., 2017) inspire continual learners such as the FlyModel (Shen et al., 2021), sparse distributed memory (Bricken et al., 2023) and DG-gated mixtures (Kapoor et al., 2026). We tested this family as a front-end and found no benefit (Section 5).

**Biology of local sleep.** Sleep-like off-periods occur locally in awake, sleep-deprived animals (Vyazovskiy et al., 2011). Causal evidence is recent: optogenetically inducing ON/OFF alternation in one cortical hemisphere of awake mice locally discharges sleep pressure and weakens synaptic markers (GluA1, pS845), bilateral induction during sleep deprivation rescues memory consolidation, and an equal *tonic* reduction of firing does not discharge the pressure (Driessen et al., 2026). Our refractory rotation is the algorithmic counterpart of this alternation requirement, and our soft-rotation control (a reduction of gain rather than an off-period) the counterpart of tonic reduction.

### 3 Experiment Setup

**Data and protocol.** Sequential axis: split-MNIST (five tasks of two classes, 5 epochs per task) and split CIFAR-10 (same protocol;  $32 \times 32$  luminance so that distance-dependent connectivity retains its geometry). Static axis: the identical learner and budgets on the i.i.d. shuffle. Waking batches have  $n_w = 16$  samples; the episodic buffer holds  $K \in \{200, 1000\}$  raw samples under reservoir writing. Forgetting  $F$  is the mean drop of each earlier task’s accuracy from its own end to the end of training (per-task matrices in Appendix I); evaluation runs with the suppression mask off ( $k$ -WTA only), so test order and batch size cannot affect it. Configurations were selected on the official test splits, which served as the development set, and then frozen. Every main-text comparison re-trains them with a stratified tenth of the training data removed from the stream and evaluates on that tenth, which took part in no selection; a second tenth, drawn with a different seed and sharing 9% of its samples with the first, replicates the headline rows (Appendix I). Results not re-run under this protocol are labelled development-phase. All numbers are mean  $\pm$  s.d. over three seeds unless marked otherwise. Runs are deterministic at a fixed CPU thread count; because the reduction order feeds  $k$ -WTA’s discontinuity, a different thread count is effectively a different seed. A third regime uses a frozen V1-like feature front-end for CIFAR-10: 256 normalised random  $6 \times 6 \times 3$  patches sampled from the training images act as fixed filters (stride 2, rectification against the per-filter mean response, quadrant average pooling; 1024-D, linear probe  $\approx 49\%$ ). Nothing in the front-end is learned; the frozen representation serves as the model input. Section 7 additionally uses split CIFAR-100 (ten tasks of ten classes, same protocol and budgets, luminance and feature variants). With tenfold fewer supervision events per class it probes the learning rule far outside the regime its constants were tuned in, and its absolute numbers are at the floor for every method tested (chance 1%, BP+ER 6–10%); we use it for orderings and mechanisms, not as a competitive benchmark.

**Architecture.**  $d$ –512–256– $C$  with  $k$ -WTA sparsity 10% on the hidden layers (the fraction selected in Section 6.3; appendix results at the earlier 5% fraction are marked as such), 30% distance-dependent connectivity on hidden synapses, Dale’s law with 80% excitatory units, and the single-sweep learning rule of Section 4. The offline-night reference uses its own preferred narrower core (256–128).

**Baselines.** (i) Backpropagation with experience replay at equal buffer  $K$  (included in every table); (ii) the offline *night*: after each waking epoch, 20 replay batches of 256 samples at  $3 \times$  learning rate with no concurrent input (the strongest schedule in our sweep); (iii) interleaved ER for the local learner; (iv) no buffer.

**Metrics.** Final accuracy, forgetting  $F$ , replay cost  $R$  (batches and samples), the isolated-channel width  $\rho_\ell$  (fraction of replay-activated units in layer  $\ell$  that are asleep for the current input), code overlap (mean pairwise Jaccard of task-active unit sets), participation-ratio dimensionality, and an idealised 45-nm FP32 arithmetic-energy proxy (Horowitz, 2014): 0.9 pJ per FP32 accumulate (one per synaptic event) for spiking inference vs. 4.6 pJ per multiply–accumulate for dense rate inference, after rate-to-spike conversion with thresholds calibrated on training images and  $T_s$  integration steps (Rueckauer et al., 2017). The proxy counts arithmetic only, with no data movement, membrane updates or comparisons, so it is a relative measure rather than a hardware figure (a fabricated neuromorphic chip reports  $\approx 26$  pJ per synaptic event (Merolla et al., 2014)).

## 4 Methodology

### 4.1 Cortical base learner

Layer activities are  $a^0 = x$  and, for  $\ell = 1, \dots, L$ ,

$$z^\ell = W^\ell a^{\ell-1} + b^\ell, \quad a^\ell = \Phi_k(\sigma(z^\ell) \odot (1 - s^\ell)), \quad (3)$$

where  $\sigma$  is a rectifier,  $s^\ell \in \{0, 1\}^{n_\ell}$  is the (possibly empty) *suppression mask* of Section 4.3, and  $\Phi_k$  keeps the  $k$  largest entries of each row and zeroes the rest ( $k$ -WTA); a per-unit gain exists in the implementation for homeostatic scaling but is fixed at 1 throughout. Layers  $1, \dots, L-1$  are hidden; layer  $L$  is a linear  $C$ -way readout whose pre-activation  $z^L$  is the output, scored by squared error against the one-hot target  $y$ . Errors are produced by one top-down sweep,

$$\varepsilon^L = y - z^L, \quad \varepsilon^\ell = \sigma'(z^\ell) \odot \left( (B^\ell)^\top [\kappa \tanh(\varepsilon^{\ell+1}/\kappa) \odot \mathbf{1}(a^{\ell+1} > 0)] \right), \quad (4)$$

i.e. a bounded ( $|\varepsilon_i^\ell| \leq \kappa \sum_j |B_{ji}^\ell|$  per component), event-gated burst signal carried by learned feedback weights  $B^\ell \in \mathbb{R}^{n_{\ell+1} \times n_\ell}$  (no transport), in the spirit of predictive-coding and burst-dependent learners with local plasticity (Whittington and Bogacz, 2017; Payeur et al., 2021). Forward and feedback weights follow the same local product with a shared decay  $\lambda$  (Kolen–Pollack alignment (Kolen and Pollack, 1994; Akrouf et al., 2019)),

$$\Delta W^\ell \propto \varepsilon^\ell (a^{\ell-1})^\top - \lambda W^\ell, \quad \Delta B^{\ell-1} \propto \varepsilon^\ell (a^{\ell-1})^\top - \lambda B^{\ell-1}, \quad (5)$$

with sign-concordant feedback under Dale’s law. The default optimiser is heavy-ball SGD ( $\mu = 0.9$ , one per-synapse velocity, no second-moment state); the  $\epsilon$ -floored masked Adam of the earlier 5% phases appears as a control (Section 5.3) and in the appendices. Equations (3)–(5) were fixed by earlier ablation ladders and are not varied here, with one exception: Section 7 replaces the constant  $\lambda$  by an adaptive per-layer controller.

## 4.2 Exactly isolated replay

Let  $X = \{x_b\}_{b=1}^m$  be the current waking batch,  $a_{i,b}^\ell$  the activity of unit  $i$  on sample  $b$ , and  $\mathcal{A}^\ell(X) = \{i : \exists b, a_{i,b}^\ell \neq 0\}$  the *awake set*. During the same forward step, a replay batch drawn from the buffer is applied through the synapse mask

$$M_{ij}^\ell = \mathbf{1}[j \notin \mathcal{A}^{\ell-1}(X) \vee i \notin \mathcal{A}^\ell(X)], \quad (6)$$

i.e. a synapse may take the replay update iff its presynaptic unit is silent for the current input or its postsynaptic unit is asleep. The mask applies to the *entire* replay-induced parameter change, not only to the data gradient: with heavy-ball velocity  $v$  ( $\mu = 0.9$ ), the local update direction  $G = \varepsilon^\ell (a^{\ell-1})^\top$  of (5) and the per-step decay coefficient  $\lambda$ , a replay step is

$$v \leftarrow M \odot (\mu v + G) + (1 - M) \odot v, \quad W \leftarrow W + \eta M \odot v - \lambda M \odot W, \quad (7)$$

so the velocity is frozen (not zeroed) outside the mask, the decay is confined to it, and biases follow the unit mask  $\mathbf{1}[i \notin \mathcal{A}^\ell(X)]$ ; under Adam the first and second moments are confined the same way. As McKee et al. (2026) caution and Bricken et al. (2023) analyse for sparse learners, unmasked optimiser state would otherwise re-inject the isolated update into awake synapses and void the guarantees below; the accuracy consequences of this choice are examined in Section 6. The readout row and its bias remain plastic, since old classes must stay calibrated; the guarantees below concern the hidden computation. Within one waking step the order is: forward pass on  $X$  under the current suppression, unmasked waking update, awake sets read from that forward pass, masked replay update on the updated weights (Algorithm 1, Appendix A). The replay micro-batch is inferred with the suppression removed, so a refractory unit may fire on replay and learn from it, and a burst of several micro-batches reuses the mask of its waking batch. Throughout, “during inference” means between the steps of the stream: the three operations run in sequence within each step, and we claim nothing about asynchronous execution or serving latency.

**Proposition 1** (Pre-silent updates are exactly invisible). *Fix  $X$  and let  $\widehat{W}^\ell = W^\ell + \Delta^\ell$  with  $\Delta_{ij}^\ell = 0$  whenever  $j \in \mathcal{A}^{\ell-1}(X)$ . Then every hidden activity on every sample of  $X$  is unchanged for any magnitude of  $\Delta$ ; if the readout row satisfies the same condition, or is held fixed, the network output is unchanged as well.*

*Proof.* By induction on  $\ell$ , for every sample  $b$ . Assume  $a_{\cdot,b}^{\ell-1}$  is unchanged (true for  $\ell = 1$  since  $a_{\cdot,b}^0 = x_b$ ). For any unit  $i$ ,  $\widehat{z}_{i,b}^\ell - z_{i,b}^\ell = \sum_j \Delta_{ij}^\ell a_{j,b}^{\ell-1}$ , and each term vanishes: if  $j \in \mathcal{A}^{\ell-1}(X)$  then  $\Delta_{ij}^\ell = 0$ ; otherwise  $a_{j,b}^{\ell-1} = 0$  for every  $b$ . Hence  $z_{i,b}^\ell$ , and therefore  $a_{i,b}^\ell$  through (3), is unchanged; the induction closes at the last hidden layer, and at the output layer whenever the readout row satisfies the same condition.  $\square$

**Proposition 2** (Post-asleep updates are invisible under a margin condition). *Let  $\Delta^\ell$  and  $\Delta b^\ell$  additionally move synapses and biases of units  $i \notin \mathcal{A}^\ell(X)$  with active presynaptic partners, and write  $\delta_{i,b} = \sum_j \Delta_{ij}^\ell a_{j,b}^{\ell-1} + \Delta b_i^\ell$ . If for every such unit and every sample  $\sigma(z_{i,b}^\ell + \delta_{i,b}) < \tau_{k,b}^\ell$ , where  $\tau_{k,b}^\ell$  is the  $k$ -th largest value entering  $\Phi_k$  on sample  $b$  (strict, so a tie cannot promote the unit; when fewer than  $k$  entries are positive,  $\tau_{k,b}^\ell = 0$  and the condition reads  $\sigma(\cdot) = 0$ ), the hidden activities on  $X$  are unchanged.*

*Proof.* A unit outside  $\mathcal{A}^\ell(X)$  contributes  $a_{i,b}^\ell = 0$  on every sample. After the update its pre-activation on sample  $b$  is  $z_{i,b}^\ell + \delta_{i,b}$ ; under the stated margin it is still excluded by  $\Phi_k$  (or rectified to zero), so  $a_{i,b}^\ell = 0$  still. The sample’s original winners keep their pre-activations, because their synapses from active presynaptic units lie outside  $\Delta^\ell$  by the mask and their silent-presynaptic terms vanish as in Proposition 1; so the same  $k$  units win with the same values, every other unit’s input is untouched, and the layer’s activities are identical. When fewer than  $k$  units are positive on sample  $b$ ,  $\tau_{k,b}^\ell = 0$  and the condition reads  $\sigma(z_{i,b}^\ell + \delta_{i,b}) = 0$ : a zero-valued entry carries no activity whether or not  $\Phi_k$  selects it.  $\square$

**Corollary 1** (Suppressed units are exact for any magnitude). *If  $i$  is suppressed by the rotation mask ( $s_i^\ell = 1$  in (3)), then  $a_{i,b}^\ell = 0$  for every sample and Proposition 2 holds with no margin condition.*

The propositions take the mask and the weights it protects from the same state. In the implemented order the awake sets precede the waking update, so they hold exactly for the pre-update weights; the one-step lag is part of the measured drift below and accounts for 0.02 of its 0.32 hidden-code points.

**Remark 1** (How exact, measured). *The default system keeps the readout plastic and does not enforce the margin of Proposition 2, so exactness beyond the proven channels is an empirical question about the training-time computation under the suppression mask (evaluation runs with the mask off). We measured it on the default configuration under the reported protocol by re-inferring every waking batch, under the suppression it was inferred with, after each of its 16,885 isolated replay updates. The update altered the top hidden code of 0.33% of the waking samples per step (three seeds, 0.31–0.35) and changed a waking prediction for 0.32%, almost all of it through the plastic readout: with the readout isolated as well, the prediction rate is 0.001% and the hidden-code rate 0.21% (one seed). The margin condition was violated for  $\sim 10^{-4}$  of the asleep units per step, and the mean maximal logit change per update was 0.02 (0.001 with the readout isolated) on the one-hot scale. On raw CIFAR-10 (one seed, development run) the hidden-code rate is 0.46% while the prediction rate rises to 3.6%, because low-margin predictions flip more easily through the plastic readout. The rates are taken under the implemented order of Algorithm 1 (Appendix A), in which the awake sets come from the activities before the waking update; masking on awake sets re-inferred after that update instead lowers the hidden-code rate only from 0.32% to 0.30% and leaves the prediction rate, the margin-violation rate and the accuracy unchanged (one seed, development run), so the residual drift comes from the unenforced margin rather than from the ordering. These rates concern the hidden code of the current batch; what replay does to later inputs is measured by the ablation of Section 6. The isolation is thus exact for silent-presynaptic and suppressed units and near-exact elsewhere, and Corollary 1 is why the refractory rotation makes it robust rather than merely approximate.*

### 4.3 Refractory rotation and its gates

**Definition 1** (Refractory rotation). *After processing waking batch  $X_t$ , set  $s_i^\ell(t+1) = \mathbf{1}[i \in \mathcal{A}^\ell(X_t)]$ : every unit that fired sits out the next competition and is therefore asleep, and consolidable, for batch  $X_{t+1}$ .*

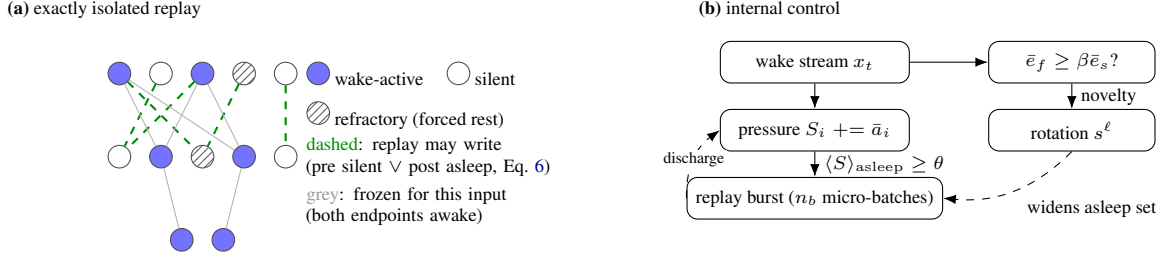
Rotation widens the isolated channel  $\rho_\ell = |\tilde{\mathcal{A}}^\ell \setminus \mathcal{A}^\ell(X)|/|\tilde{\mathcal{A}}^\ell|$ , the fraction of the units  $\tilde{\mathcal{A}}^\ell$  activated by the replay micro-batch that are asleep for the current input, from 0.2–0.3 (natural silence) to  $\approx 0.53$ , at the price of running inference on the second-best coalition. Two internal signals control the system (Figure 1):

**Homeostatic pressure (when to replay).** Each unit integrates its own use,  $S_i \leftarrow S_i + \bar{a}_i(X_t)$  with  $\bar{a}_i(X_t)$  the fraction of the batch’s samples on which unit  $i$  fired, and a replay *burst* of  $n_b$  consecutive micro-batches (one mask for all of them) is triggered when the mean pressure over the currently asleep units of all hidden layers exceeds  $\theta$ ; consolidation discharges the pressure of the units that received it,  $S_i \leftarrow (1 - \delta \mathbf{1}[i \text{ asleep}]) S_i$  with  $\delta = 1$  throughout. This is a unit-level analogue inspired by the homeostatic Process S of sleep regulation (Borbély, 1982).

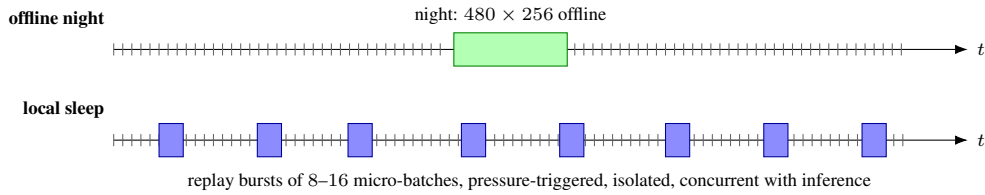
**Relative novelty (when to rotate).** With fast and slow EMAs of the batch-mean squared output error  $e_t$ ,  $\bar{e}_f(t) = (1 - \alpha_f)\bar{e}_f + \alpha_f e_t$ ,  $\alpha_f = 0.1$ , and  $\bar{e}_s$ ,  $\alpha_s = 0.002$ , rotation runs only while

$$\bar{e}_f(t) \geq \beta \bar{e}_s(t), \quad \beta \approx 1.1\text{--}1.5, \quad (8)$$

i.e. while the stream is *new relative to the learner’s own recent history* (loosely inspired by cholinergic novelty signalling (Hasselmo, 2006)). Monitoring streaming error with exponentially weighted averages is standard in concept-drift detection (Ross et al., 2012), and related loss-ratio triggers appear in concurrent continual-learning work. The detector itself is therefore not new; what is new is what it gates (the rotation, and hence the width of the replay channel) and the unmastered clause below. A single absolute threshold cannot serve across the regimes considered here: the converged error of a ten-class i.i.d. stream exceeds any level a two-class task dips under, and we observe exactly this failure (gate open 100% of the time on static MNIST, 95% on CIFAR, for any threshold that behaves on split-MNIST). Because



**Figure 1: Mechanism.** (a) During each waking batch, replay from the episodic buffer is written only into synapses whose update is invisible to the current input, exactly so for silent presynaptic and suppressed units (Propositions 1–2); refractory rotation forces recently used units into the consolidable set (Corollary 1). (b) Two internal signals: unit-level homeostatic pressure triggers replay bursts, and a relative-novelty gate decides when rotation runs. The default system runs rotation always on and replays one micro-batch at every waking batch; the pressure trigger and the novelty gate are the schedulers studied under sparse replay budgets and lower activity fractions.

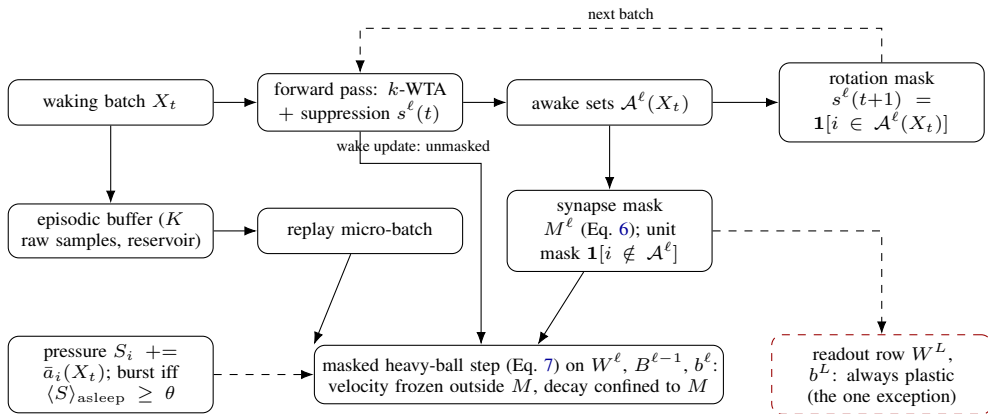


**Figure 2: Consolidation schedules.** Ticks are waking batches. The offline night pauses the stream; local sleep consolidates during it, in replay bursts confined to asleep units.

rotation is *beneficial* on streams the learner never masters (Section 5), the final gate adds an “unmastered” clause: with  $e_0$  the mean error over the first 20 batches (a chance-level estimate), rotation runs while

$$\bar{e}_f(t) \geq \beta \bar{e}_s(t) \quad \text{or} \quad \bar{e}_f(t) \geq \gamma e_0, \quad \gamma \approx 0.5. \quad (9)$$

The first clause opens the gate at task switches, the second keeps it open on any stream whose error remains a substantial fraction of chance; both are scale-free.



**Figure 3: Training data flow.** Each waking batch is inferred under the current suppression mask; its awake sets define the synapse and unit masks for the replay micro-batch applied in the same step and the rotation mask for the next batch. The masked step confines the velocity and the decay to the mask; the readout row is the one deliberately plastic exception. Unit-level pressure decides when replay bursts run.

#### 4.4 Cost accounting

**Updates and wall-clock.** Samples are not the only cost. The headline schedule makes 16,885 masked updates against the night’s 480 (35×; counts from the reported runs, whose stream is nine tenths of the training set), and its training wall-clock on one CPU thread is 8.8 min against 1.4 for the night, 1.0 with no replay and 8.6 for unmasked interleaved

replay (one run each, same protocol and thread count). The per-batch replay step, not the sample count, is the online price of consolidating during the stream, in an unoptimised implementation whose mask construction is dense.

**Baseline tuning and paired comparison.** The offline-night reference was swept over replay batch count (20, 40 per night), replay batch size (16, 64, 256), learning-rate gain ( $3\times$ ), core width (256–128 and 512–256), buffer size (200, 1000, 5000) and timing (clocked after each epoch, novelty-timed, surprise-timed), and its best sequential cell is reported. BP+ER was swept over learning rate ( $3\cdot 10^{-4}$ ,  $10^{-3}$ ,  $3\cdot 10^{-3}$ ), width and buffer size, and the local learner over  $\eta$  (Appendix F), replay batch size, cadence and the gates of Table 1. Selection everywhere was by sequential accuracy at  $K=1000$  on the official test split (the development set), with the static axis reported alongside; the comparisons reported are on the held-out split (Appendix I). Pairing seeds by index on that split, the local system’s margin over the narrow-substrate night is +2.5 points (bootstrap 95% CI [+1.3, +4.8], three seed pairs), over the same-substrate night +2.6 ([+0.2, +5.2], six pairs; the night’s seed variance makes this one fragile, and on the second split it shrinks to +0.3), over unmasked interleaved ER +6.3 ([+3.8, +9.2]), over BP+ER +2.8 ([+2.5, +3.2]), over the silent mask +3.6 ([+3.1, +4.1]) and over rotation alone +0.1 ([−0.3, +0.5]). Replay cost is measured in samples ( $R_s$  = batches  $\times$  batch size) and synaptic work. The night costs  $R_s = 480 \times 256 \approx 1.2 \times 10^5$  samples. The headline configuration (one 16-sample micro-batch per waking batch,  $91.6 \pm 0.3$ ) costs  $16,885 \times 16 \approx 2.7 \times 10^5$  samples ( $2.2\times$  the night). Two schedules meet the night’s budget within  $1.1\times$ : eight-sample micro-batches at every waking batch ( $1.35 \times 10^5$ ,  $91.1 \pm 0.4$ ) and 16-sample micro-batches at every second one (8,442 updates,  $1.35 \times 10^5$ ,  $91.2 \pm 0.2$  sequential,  $93.1 \pm 0.2$  static); pressure-triggered bursts of 16 at a third of the events reach  $88.6 \pm 1.6$ . Accuracy therefore depends on the frequency of small isolated updates rather than on the number of replayed samples.

## 5 Results

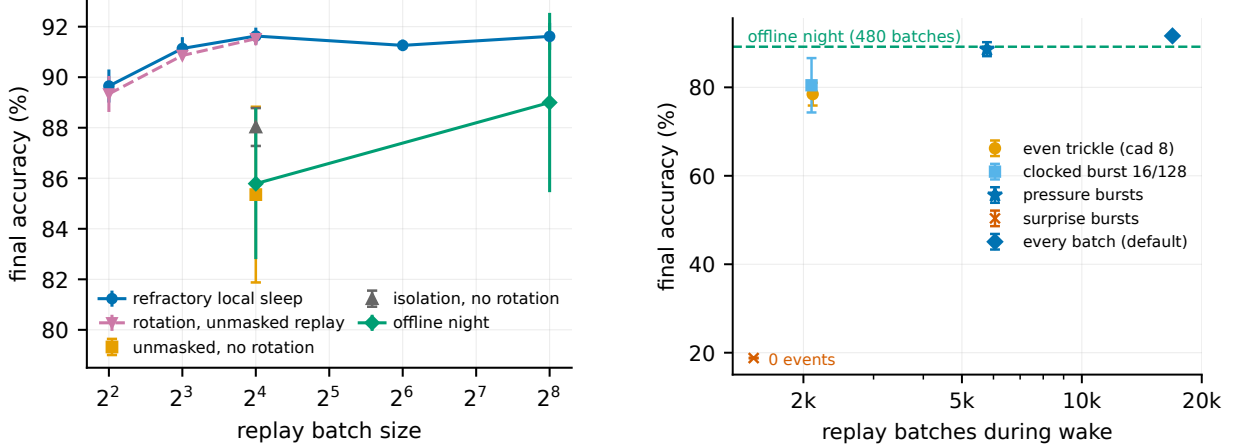
### 5.1 Local sleep replaces the night

On split-MNIST at  $K=1000$ , refractory local sleep with an isolated replay micro-batch beside every waking batch reaches  $91.6 \pm 0.3\%$  (six seeds) with no offline phase. This is above the best offline night ( $89.2 \pm 1.7\%$  on the night’s preferred narrow substrate;  $89.0 \pm 3.5$ , highly variable across seeds, on ours; paired margins +2.5 [+1.3, +4.8] and +2.6 [+0.2, +5.2]), above unmasked interleaved ER at the same budget ( $87.5 \pm 0.8$  under Adam,  $85.4 \pm 3.5$  under SGD) and above BP+ER ( $88.8 \pm 0.3$ ; +2.8 [+2.5, +3.2]). In a single pass over the stream (one epoch per task, three seeds) the ordering sharpens:  $91.8 \pm 0.2$  against  $88.7 \pm 0.3$  for BP+ER,  $76.0 \pm 20.1$  for unmasked ER and  $75.1 \pm 3.5$  for the offline night, for which one night per task is too few (Table 1). Halving the replay cadence with 16-sample batches costs 0.4 ( $91.2 \pm 0.2$ ). Adding a night on top of continuous local sleep adds 0.8 ( $92.4 \pm 0.3$  vs.  $91.6 \pm 0.3$ ): once consolidation runs during wake, the offline phase is nearly redundant at this buffer size. At very small buffers ( $K=200$ ) the night retains an edge only in combination with narrow substrates (Appendix D).

**A second split.** On a second, independent held-out tenth the ordering holds (rotation  $91.8 \pm 0.4$ , BP+ER  $88.5 \pm 0.7$ , unmasked ER  $80.0 \pm 14.1$ , the narrow night  $89.3 \pm 0.6$ , the silent mask  $88.6 \pm 1.3$ , rotation alone  $91.8 \pm 0.6$ ), with one exception: the same-substrate night, bimodal on the first split, reaches  $91.5 \pm 0.5$  on the second and trails by only 0.3 there (Appendix I).

### 5.2 What rotation and isolation each buy

Figure 4 (left) varies the replay batch size. The full system is flat from 256 down to 8 samples ( $91.6 \pm 0.5 \rightarrow 91.1 \pm 0.4$ ) and loses 1.9 at 4 ( $89.7 \pm 0.7$ ). At batch 16, unmasked replay without rotation reaches  $85.4 \pm 3.5$ , isolation without rotation  $88.0 \pm 0.7$ , and the offline night falls from  $89.0$  to  $85.8 \pm 3.0$  when its batches shrink to 16. The fourth cell of the square, rotation with unmasked replay, reaches  $91.5 \pm 0.3$  at batch 16,  $90.9 \pm 0.1$  at 8 and  $89.3 \pm 0.7$  at 4. The two factors are therefore strongly sub-additive: rotation is the larger single factor (+6.1 alone, +3.6 on top of isolation), and isolation’s seed-paired margin is 0.1 [−0.3, +0.5] at batch 16, 0.3 [−0.2, +0.6] at 8 and 0.3 [−0.5, +1.2] at 4, within seed noise at every size, with 0.4 on the static axis ( $94.1$  vs.  $93.7$ ). At two-sample batches both systems lose heavily, but differently. Isolated replay degrades gracefully ( $79.0 \pm 3.2$ , six seeds within 75–84), whereas unmasked replay bifurcates: three seeds finish at 82–85, one at 58 and two at chance, i.e. severe failures below 60% in half the seeds (Appendix H). In the seeds and configurations tested, isolation’s accuracy value is thus the absence of those failures rather than a margin. What it buys at every batch size is the guarantee that, for the current input under its suppression mask, replay leaves the hidden activities unchanged, exactly on the proven channels and within the margin elsewhere. Without it every micro-batch perturbs the live coalition, and the outcome rests on the rotation’s tolerance of that perturbation, which runs out at batch 2.



**Figure 4: Left (isolation):** final split-MNIST accuracy vs. replay batch size (default substrate, heavy-ball SGD); the number of updates is fixed within each series (one per waking batch; 20 per epoch for the night, whose samples therefore fall with its batch). The full system is flat down to 8 samples and loses 1.9 at 4 (12.7 at 2, Appendix H); rotation with unmasked replay (dashed) trails it by 0.1 to 0.3, within seed noise, and fails severely in half the seeds at batch 2; without rotation, unmasked and isolated replay and the offline night all degrade. **Right (timing):** accuracy vs. number of waking replay batches. Trickles and clocked bursts fail alike; pressure-triggered bursts (star) keep most of the accuracy at a third of the events; surprise triggering fires no events in this regime.

### 5.3 The direction of protection matters, and the least optimiser state suffices

Three controls sharpen the construction. First, *mirror isolation* implements the classical protected direction with our own machinery: the *waking* update is barred from synapses whose endpoints both served the last replay batch (protecting the past), while replay itself is applied unmasked. With everything else identical, mirror isolation reaches only  $78.0 \pm 3.2\%$ , below even unmasked interleaved ER ( $87.5 \pm 0.8$  under Adam) and far below our direction ( $91.6 \pm 0.3$ ). Constraining wake learning impairs the new task while the unmasked replay keeps perturbing the live computation, a double loss. The reversed mask thus costs fourteen points (20, with high seed variance,  $70.3 \pm 10.1$ , at the 5% fraction in development). It is one implementation of the protect-the-past direction, not a test of the methods in that literature. Second, replacing masked Adam with heavy-ball SGD (a single per-synapse velocity and no second-moment state) inside the same isolation gives  $91.6 \pm 0.3\%$  sequential and  $94.1 \pm 0.2\%$  static (six seeds), a tie with masked Adam ( $91.5 \pm 0.4/94.1 \pm 0.1$ ; unmasked SGD control  $85.4 \pm 3.5$ ). Carrying a single velocity and nothing else is simpler at equal accuracy and closer to biology; carrying *nothing* is not. With momentum zero the isolated replay micro-batches lose the velocity’s noise averaging, activity collapses to chance at every learning rate unless the replay step is re-scaled by hand, and the best pure-SGD setting then trails by 3.2 sequential and 4.7 static points ( $88.4 \pm 0.5/89.4 \pm 0.3$ ). On the held-out split, confined and leaking moments tie (Section 6), so confining the update over time costs nothing. Third, *soft* rotation (halving, rather than silencing, recent winners) loses on both axes ( $89.9 \pm 0.7$  sequential,  $84.3 \pm 1.0$  static): all-or-none OFF periods beat turned-down gain, echoing the biological finding that tonic firing-rate reduction does not discharge local sleep pressure while ON/OFF alternation does (Driessen et al., 2026). Table 1 in Section 6 collects these controls together with the remaining component knockouts. A two-sided learning-rate sweep (Appendix F) shows that the optimiser comparison is a plateau rather than a knife-edge, with masked Adam at its own best rate.

### 5.4 When replay must be rare: homeostatic bursts, not surprise

Figure 4 (right): at 12.5% of the replay events, a clocked burst of 16 batches every 128 reaches  $80.5 \pm 6.1$ , no better than an even trickle at the same budget ( $78.5 \pm 2.6$ ). Unit-level pressure triggering keeps  $88.6 \pm 1.6$  at a third of the events. Surprise triggering fails: at the 10% fraction the converged error never crosses the surprise threshold, so *no* replay event fires (18.8, the forgetting floor), and at the 5% fraction, where it fired occasionally, it still failed at every volume. Homeostatic triggering therefore outperforms the tested surprise trigger during wake, the opposite of the night, whose best trigger in our sweeps is novelty-timed.

### 5.5 The i.i.d. price of rotation

Rotation’s i.i.d. price depends on the activity fraction. At the 5% fraction it cost 1.5 points on i.i.d. MNIST (92.6 vs. 94.1 rotation-free; 1.3–3.0 across the development-phase gate grids) and, in development, *reversed* into a +2.5 to

+6.5-point static gain on split CIFAR-10, where forced rotation acts as a regulariser in the underfit regime. At the 10% fraction the MNIST price is 1.2–1.7 points (94.1 vs. 95.3–95.8 rotation-free) and the CIFAR static effect is inconclusive under the available seeds ( $28.0 \pm 1.2$  vs.  $30.0 \pm 1.9$ ). Under masked Adam, a relative-novelty gate with the unmastered clause (Eqs. (8)–(9)) removed the 5% price with one setting across all four regimes; the full gate study, including why a single absolute threshold cannot serve across these regimes and why unit-level pressure gating only trades the axes, is in Appendix C. Under heavy-ball SGD the gate had to be committed to bouts (Section 6). At the 10% fraction both remedies become largely unnecessary, because the activity fraction itself (Section 6.3) shrinks the price at its source.

## 5.6 Transfer to CIFAR-10

On split CIFAR-10 (grayscale, same protocol) the ordering transfers and widens: refractory local sleep  $28.9 \pm 1.0 >$  offline night  $25.7 \pm 1.2 >$  BP+ER  $24.8 \pm 0.3 >$  unmasked ER  $15.4 \pm 4.2 \approx$  no buffer 16.3 (the same on the second split:  $29.7 > 26.5 > 25.8 > 15.5$ ), despite task-active code overlap of 0.9–0.98. On the V1-like feature front-end the local ordering persists (refractory  $42.3 \pm 0.6 >$  offline night  $34.0 \pm 0.8 \gg$  unmasked  $20.5 \pm 1.0$ ), but BP+ER reaches  $42.8 \pm 0.7$  at its best learning rate, the reverse of the raw-pixel ordering. At the 5% fraction the gap was 4.7 points (local learner at 36.8, development phase); tuning both sides and raising the activity fraction reduce it to 0.5 ( $42.3 \pm 0.6$  local vs.  $42.8 \pm 0.7$  BP+ER). Most of the gap was therefore the local learner’s optimiser and its starved activity fraction, not its learning rule. On raw pixels the local learner leads outright.

The residual gap is consistent with the *substrate*, not the learning rule. Transplanting the local network’s ingredients into the BP net one at a time (each cell at its better of two learning rates): its width *helps* BP+ER ( $43.7 \pm 1.0$  for a dense 512–256 net); its 30% distance-dependent wiring, drawn by the same generator on the same sheets, is neutral ( $43.5 \pm 0.2$ ); and its  $k$ -WTA costs 3 points ( $40.4 \pm 1.5$  at 10%;  $39.5 \pm 0.6$  at 5%), leaving backprop-with-replay *below* the local learner on the identical substrate ( $42.3 \pm 0.6$  vs.  $40.4 \pm 1.5$  at 10%; three seeds each). What separates the two systems on feature inputs is dense activation, not backpropagation’s credit assignment: on the substrate used here, the decomposition leaves no gap attributable to the learning rule. The advantage over *other local schedules* (nights, unmasked ER) holds across all three input regimes.

## 5.7 Energy

Converted to spiking inference ( $T_s=8$ ; firing thresholds calibrated on training images, evaluated on the held-out tenth), the default system (always-on rotation, heavy-ball SGD, 10% fraction; rate accuracy 94.2% on the measured seed) runs at 92.7% and 117 nJ/sample vs. 2461 nJ for dense rate inference ( $21\times$ ) and 526 nJ for event-driven rate inference ( $4.5\times$ ), improving with the time budget (94.0% at  $T_s=16$ , 236 nJ; 94.0% at  $T_s=32$ ). The rotation-free control converts to 94.2% at  $T_s=8$  and dips slightly to 93.9% at  $T_s=32$ ; the interaction between rotation and spike calibration is therefore small under heavy-ball SGD on the held-out split.

## 5.8 Negative results

The following alternatives were tested under matched budgets and rejected: a frozen dentate-gyrus-style sparse expansion front-end (input decorrelation does not propagate through learned  $k$ -WTA layers: task overlap at the input drops  $0.79 \rightarrow 0.18$ , yet every downstream metric worsens by 3–11 points); generative (REM) replay as a substitute for episodic samples; margin-based reverse learning; surprise-gated memory *writing*; multiplicative burst coding; absolute-threshold rotation gates; pruning anchored to sparse replay bursts (Appendix E); soft (gain-reduction) rotation and mirror-direction isolation (Section 5.3); an OFF-side gate refractory, the inactive half of the flip-flop in this system; strongly coupled synaptic anchors; and utility-exempt rotation as a route past the bout gate (all Section 6).

## 6 Ablation Study

The system under test combines five ingredients: a biologically constrained substrate, the isolation mask (6), the refractory rotation, a heavy-ball optimiser, and the gate (9). This section removes or replaces one ingredient of the consolidation mechanism at a time (Table 1). The substrate beneath it was fixed by earlier ablation ladders; its cost accounting (1.2 points below a dense backpropagation MLP at  $4\times$  fewer synaptic operations per pass) is reproduced in Appendix B. Every mechanism cell shares the default substrate (512–256 hidden, 10%  $k$ -WTA, 30% distance-dependent connectivity), the same buffer ( $K=1000$ , random writing) and the same replay budget (waking batch 16, replay batch 16 after every waking batch). The appendices report the corresponding grids at the 5% fraction, where each ordering below was first established.

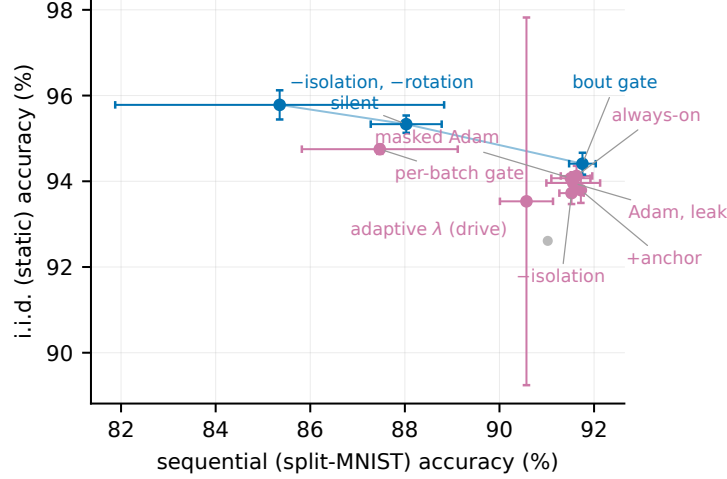
**Table 1: Component ablation of the consolidation mechanism** on the default substrate (512–256/10%; backprop references dense 256–128). Sequential = final accuracy on split-MNIST; static = i.i.d. MNIST (the axis that must not degrade). Dashes: combination not applicable; blank: not run. Single pass = one epoch per task, otherwise identical. Held-out split; headline rows carry six seeds, the rest three.

| Variant                                                                     | Sequential        | i.i.d. (static) | Seq., single pass |
|-----------------------------------------------------------------------------|-------------------|-----------------|-------------------|
| <b>full system:</b> isolated replay + always-on rotation (heavy-ball SGD)   | <b>91.6 ± 0.3</b> | 94.1 ± 0.2      | 91.8 ± 0.2        |
| rotation gated by relative novelty, committed to bouts                      | 91.8 ± 0.3        | 94.4 ± 0.3      |                   |
| rotation gated per batch                                                    | 87.5 ± 1.6        | 94.7 ± 0.1      |                   |
| + two-timescale anchor ( $\lambda=\mu=3\cdot 10^{-4}$ ), rotation always on | 91.7 ± 0.0        | 93.8 ± 0.3      |                   |
| – rotation (same mask, Eq. 6, natural silence only)                         | 88.0 ± 0.7        | 95.3 ± 0.2      |                   |
| – isolation, – rotation (unmasked interleaved replay)                       | 85.4 ± 3.5        | 95.8 ± 0.3      | 76.0 ± 20.1       |
| – isolation, + rotation (rotation alone)                                    | 91.5 ± 0.3        | 93.7 ± 0.3      |                   |
| readout-only replay (hidden layers take no replay update)                   | 41.7 ± 10.9       | 82.7 ± 0.3      |                   |
| random rotation (matched suppression count)                                 | 91.0 ± 0.8        | 92.1 ± 0.2      |                   |
| – replay (no buffer)                                                        | 19.4 ± 0.0        | 95.1 ± 0.0      | 18.1 ± 0.7        |
| masked Adam, moments confined to the mask                                   | 91.5 ± 0.4        | 94.1 ± 0.1      |                   |
| masked Adam, moments leaking outside the mask                               | 91.6 ± 0.6        | 94.0 ± 0.3      |                   |
| momentum 0 (pure SGD; $\eta$ and replay gain re-tuned, else chance)         | 88.4 ± 0.5        | 89.4 ± 0.3      |                   |
| offline night instead of concurrent replay, same substrate                  | 89.0 ± 3.5        | —               | 75.1 ± 3.5        |
| the same night on its preferred narrow substrate                            | 89.2 ± 1.7        | —               |                   |
| mirror direction: protect the past                                          | 78.0 ± 3.2        | 91.7 ± 0.6      |                   |
| soft rotation: gain 0.5 instead of 0                                        | 89.9 ± 0.7        | 84.3 ± 1.0      |                   |
| unmasked ER at the same budget (Adam)                                       | 87.5 ± 0.8        | 95.1 ± 0.2      |                   |
| backprop (+ER sequential; plain static), dense                              | 88.8 ± 0.3        | 97.4 ± 0.1      | 88.7 ± 0.3        |

## 6.1 Reading the component table

The first block prices each ingredient by what its removal costs the sequential axis under heavy-ball SGD. Rotation is worth +3.6 points (91.6 vs. 88.0 for the same mask on natural silence alone), rotation and isolation together +6.2 with a twelve-fold smaller seed deviation (91.6 ± 0.3 vs. 85.4 ± 3.5), isolation alone +2.6 (88.0 vs. 85.4), isolation on top of rotation +0.1 (91.5 ± 0.3 for rotation with unmasked replay; [−0.3, +0.5] paired), and replay is worth everything (19.4 is the forgetting floor). Replay confined to the readout row, with the hidden layers frozen for replay, falls to 41.7 ± 10.9 (ten classes) and costs 11 static points: readout-only replay cannot account for the full system’s accuracy, so hidden-layer replay updates are essential to it. A random suppression of matched size in place of the refractory rule recovers most of rotation’s gain (91.0 ± 0.8) but trails it by 0.6 sequential and 2.0 static points with  $1.4\times$  the forgetting ( $F=8.7$  vs. 6.4): alternating the coalition carries most of the effect, and resting the recent winners the rest. The static column shows which component pays: rotation is the only statically costly one (94.1 always-on against 95.3–95.8 without it, and 93.7 with rotation and no isolation), a price of 1.2–1.7 points at the 10% fraction. The unmasked control is not handicapped by the replay step ( $3\eta$  against a waking step of  $\eta/16$ ). Sweeping the replay gain over  $\{1/16, 1/4, 1, 3, 10\}$  moves masked replay 67.2 → 86.8 → 90.8 → 91.6 → 74.2 and unmasked replay 61.6 → 81.4 → 85.7 → 85.4 → 47.3 (sequential; three seeds, six at gain 10, where both destabilise): masked replay peaks at the default, unmasked replay plateaus from gain 1 to 3, and the rotation-free unmasked control never closes the gap. The sweep bounds that control; isolation’s own contribution is estimated by the square above. The gates that were built to refund a larger price at the 5% fraction are neutral here: the bout-committed gate trades +0.2 sequential for +0.3 static (91.8 ± 0.3/94.4 ± 0.3), the per-batch gate is erratic (87.5 ± 1.6/94.7 ± 0.1), and the two-timescale anchor no longer adds anything (91.7 ± 0.0/93.8 ± 0.3). Their mechanisms remain instructive. At 5% the per-batch gate’s flicker destroyed heavy-ball SGD’s gain; committing the gate to bouts (loosely modelled on the consolidated bouts of the sleep–wake switch (Saper et al., 2005), which motivates commitment but not the specific minimum duration) repaired it; and bout-length and OFF-refractory controls located the residual price in the rotation of settled coalitions itself (Appendix G). At the 10% fraction, however, the larger activity fraction has absorbed most of what the gates bought, and the default system is simply always-on rotation.

The middle block tests the confinement requirement itself. Propositions 1–2 require Adam’s moments to advance only inside the mask. State advancing outside re-injects the replay gradient into awake synapses through the next unmasked waking step: the present batch’s invariance still holds, since the weight change itself stays masked, but the confinement of the update over time is lost. Empirically, on the held-out split, confined and leaking moments tie (91.5 ± 0.4 vs. 91.6 ± 0.6), as does the single velocity (91.6 ± 0.3). The velocity is kept as the least state that stays confined to the



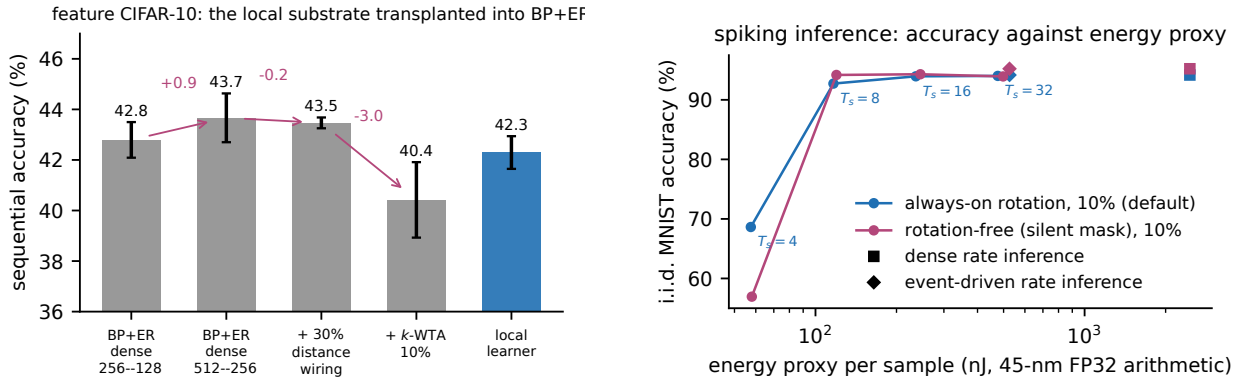
**Figure 5: The operating frontier** at the 10% fraction under heavy-ball SGD: split-MNIST sequential accuracy against i.i.d. static accuracy (headline points six seeds). Blue: non-dominated by seed means (the bout-committed gate, the silent mask, unmasked rotation-free replay); purple: dominated; held-out split. The top cluster (always-on rotation, bout gate, anchor, masked Adam) lies within seed noise of one another; the frontier trades sequential for static accuracy through the silent and unmasked variants, and the adaptive- $\lambda$  point (Section 7) is dominated on this split.

mask at equal accuracy; removing it as well (momentum zero, Table 1) costs 3.2/4.7 points once the replay step has been re-scaled to survive at all.

One budget note completes the block: half the cadence with 16-sample batches costs 0.4 ( $91.2 \pm 0.2$ ), so the system prefers consolidation to run continuously.

## 6.2 Two probes of the residual static price

At the 5% fraction, the OFF-refractory control located the bout gate’s residual static price in the rotation of settled coalitions itself. Two mechanisms addressed that diagnosis directly (details in Appendix G; 5% numbers). *Utility-exempt rotation*, which spares the top- $q$  long-use units from ever being benched, only traces a trade-off that the bout gate dominates: temporal commitment beats structural exemption. *Two-timescale synapses* in the spirit of synaptic consolidation cascades (Benna and Fusi, 2016) do not dissolve the price, but weak symmetric coupling acts as a *sequential stabiliser* at 5% ( $92.4 \pm 0.2$ , variance halved, static parity); at the 10% fraction the anchor is neutral ( $91.7 \pm 0.0/93.8 \pm 0.3$ ). Anchor and bout gate compose sub-additively. Figure 5 draws the operating frontier.



**Figure 6: Left:** the substrate decomposition on feature CIFAR-10, transplanting the local network’s ingredients into BP+ER one at a time (each cell at its better of two learning rates; three seeds). **Right:** spiking inference, i.i.d. MNIST accuracy against the idealised 45-nm FP32 arithmetic-energy proxy for  $T_s \in \{4, 8, 16, 32\}$ , with the dense (square) and event-driven (diamond) rate references; held-out split; the default is monotone in  $T_s$ , the rotation-free control dips 0.4 at  $T_s=32$ .

### 6.3 The activity fraction

The substrate’s 5%  $k$ -WTA fraction was fixed early in the project. The decomposition above suggested that it was too low: if sparse  $k$ -WTA costs BP several points on features, it presumably costs our own learner too. It does, on both regimes, and the cost is recovered almost for free; this subsection documents the sweep that led to the 10% fraction used throughout the main text. On feature CIFAR-10, relaxing the fraction improves *both* axes monotonically: sequential  $40.9 \rightarrow 42.9 \pm 0.6$  and static  $38.3 \rightarrow 46.4 \pm 1.6$  from 5 to 25%, with forgetting slightly *reduced*. The BP control traces the same curve ( $39.5 \rightarrow 40.4 \rightarrow 40.7$  at 5, 10, 15%), with the local learner 1.4–1.9 points ahead at every matched level. On MNIST the sequential axis is flat from 10% upward: always-on rotation reaches  $91.6 \pm 0.3$  sequential at  $94.1 \pm 0.2$  static (six seeds), 15% gives  $91.4 \pm 0.4/94.4 \pm 0.2$  and 25% gives  $91.5 \pm 0.4/94.6 \pm 0.2$ , against  $91.0 \pm 0.5/92.6 \pm 0.0$  at 5%.

Three controls sharpen the reading. First, the no-buffer ceiling itself rises with the fraction (94.1 at 5%, 95.1 at 10%), so rotation’s static price *at matched sparsity* shrinks from 1.5 to 1.0 points, and the need for a gate shrinks with it: at 10% always-on rotation no longer trails the bout-gated system. Second, in development, rotation-free silent isolation gained nothing from the larger fraction ( $89.0 \rightarrow 89.3 \rightarrow 88.7$ , official test set): the change helps specifically the *rotating* family, which at 5% was starved of representational room. Third, asymmetric schedules (a mild input-facing layer above a sparse deep layer) failed in development, with sequential parity to uniform relaxation but a 3–4-point static deficit: the cost is levied at every layer. The larger fraction is free elsewhere, since synaptic operations rise only 3–5%. The default substrate therefore uses 10%; Table 1 and every main-text number use it, with the complete mechanism grid re-established there and all orderings preserved, while the appendices report the 5% grids on which each mechanism was first isolated. The activity fraction, not the gate, is the cheapest remedy for rotation’s static price (Figure 5).

## 7 An adaptive decay controller

**The last tuned constant.** Every result above runs the rule of (5) with a decay  $\lambda$  tuned once per dataset ( $10^{-3}$  on the image benchmarks). Split CIFAR-100, ten tasks of ten classes, exposes what that constant hides. With one hundred classes, each class receives a tenth of the per-epoch supervision events, while the decay acts on every synapse at every step, undiminished. At the MNIST-tuned value the hidden-to-hidden weights follow the pure-decay trajectory  $(1 - \lambda)^t$  *exactly*, because the thinned error signal loses the race outright; deep activity collapses by  $5\times$ , the readout gradient goes to zero, and the network lands at chance, 2.4% i.i.d. against 20–23% for a ten-class subset of the same images on the same substrate (development-phase diagnosis). The class count of the supervision signal is a hidden variable that every fixed  $\lambda$  is silently tuned against. A manual rescue ( $\lambda=10^{-4}$ ,  $3\times$  targets) revives activity but survives the full protocol only at 1.0–1.2%.

**The controller.** Our design principle that control signals must be self-referenced rather than absolute (Section 4.3) applies at the synapse as well. We replace the constant by a per-layer ratio controller,

$$\lambda_\ell = \min\left(\lambda_{\max}, \rho \frac{\text{EMA}[u^\ell]}{|W^\ell| + \epsilon}\right), \quad \rho = 0.25, \quad \lambda_{\max} = 0.02, \quad (10)$$

where  $u^\ell$  is the mean magnitude of the data-driven, post-optimiser, *pre-decay* waking increment applied to layer  $\ell$  (so the decay never feeds its own drive), the EMA (coefficient 0.02) runs over waking, unmasked steps only,  $\epsilon = 10^{-12}$  floors the denominator, and  $W^\ell$  and  $B^{\ell-1}$  share  $\lambda_\ell$ , so Kolen–Pollack alignment is untouched. The decay’s mean magnitude per step,  $\lambda_\ell |W^\ell|$ , is held at the fraction  $\rho$  of the mean data-driven update magnitude: layers and regimes with rich supervision keep a strong regulariser, starved ones are automatically spared.  $\rho$  is the only control target;  $\lambda_{\max}$ , the EMA coefficient and  $\epsilon$  are global safeguards never tuned per dataset. Isolation and rotation decide *where* and *when* to consolidate; the controller decides *how strongly* to update as the scale changes. The same principle underlies every control signal in this paper: novelty is a fast-over-slow error ratio, homeostasis a unit’s own use, decay a drive-over-weight ratio, all referenced to the learner’s own current state and none absolute. A gradient-referenced variant ( $\rho \eta \text{EMA}[|g^\ell|]/|W^\ell|$ ) behaved equivalently in development (official test set, not re-run under the held-out protocol); we use the drive-referenced form because it is optimiser-agnostic.

**One ratio target, every dataset.** Table 2 gives the cross-dataset comparison. On the sequential axis the controller concedes 1–3 points to the per-dataset tuned value. On the static axis it gains 1.2–5.3 points on CIFAR-10, feature CIFAR-10 and both CIFAR-100 variants; on split-MNIST it gains 1.1 on the second held-out split ( $95.2 \pm 0.4$  vs.  $94.1 \pm 0.6$ ) but one of six seeds collapsed on the first ( $93.5 \pm 4.3$  vs.  $94.1 \pm 0.2$ ), so we claim no static gain there. On CIFAR-100 the controller lifts the system  $2.7\text{--}4.7\times$  over the manual rescue with no dataset-specific decay coefficient (floor-regime numbers, reported for ordering only; the BP+ER reference reaches 6.8 raw and 10.1 with features). The

**Table 2: One dataset-independent ratio target replaces per-dataset decay tuning.** Sequential / static accuracy (%) under the tuned fixed decay ( $10^{-3}$  on the image benchmarks;  $10^{-4}$  with  $3\times$  targets as the CIFAR-100 rescue) against the drive-referenced controller of (10) at the same  $\rho = 0.25$  everywhere; held-out split, three seeds (six on split-MNIST); the full system (refractory rotation, isolated replay,  $K=1000$ ) in every cell.

|                  | fixed $\lambda$ (tuned) |                | controller (drive) |                |
|------------------|-------------------------|----------------|--------------------|----------------|
|                  | seq.                    | static         | seq.               | static         |
| split-MNIST      | $91.6 \pm 0.3$          | $94.1 \pm 0.2$ | $90.6 \pm 0.6$     | $93.5 \pm 4.3$ |
| split CIFAR-10   | $28.9 \pm 1.0$          | $28.0 \pm 1.2$ | $26.2 \pm 1.1$     | $30.1 \pm 0.2$ |
| feature CIFAR-10 | $42.3 \pm 0.6$          | $42.4 \pm 1.1$ | $39.1 \pm 0.4$     | $43.6 \pm 0.6$ |
| split CIFAR-100  | $1.2 \pm 0.1$           | $1.0 \pm 0.1$  | $3.2 \pm 0.8$      | $4.1 \pm 0.6$  |
| + features       | $1.1 \pm 0.3$           | $1.1 \pm 0.2$  | $5.2 \pm 1.6$      | $6.4 \pm 0.6$  |

realised  $\lambda_\ell$ , logged throughout, differentiates exactly as the tuning history would predict:  $\sim 10^{-5}$  on MNIST sequential streams,  $\sim 10^{-4}$  on static ones, up to  $3 \times 10^{-3}$  on raw CIFAR, and layer-graded within each net. The per-dataset  $\lambda$  ladder was therefore a signal-to-decay ratio in disguise. On the operating frontier (Figure 5) the controller point is dominated by the default on the first split ( $90.6/93.5$  against  $91.6/94.1$ ): on MNIST the tuned fixed decay remains the better setting on both axes, and the controller’s value lies on harder inputs and at depth.

**Depth is the same boundary, and two mechanisms cross it.** The class count thins the error signal per class; depth attenuates it per layer. This is the same starvation on a different axis. A depth ladder (three, four and five hidden layers, 512–256–128 ..., everything else the default configuration) makes the parallel exact. Under the tuned fixed decay without skips, the static axis dies at four hidden layers ( $10.1 \pm 0.2\%$ , chance) and the sequential axis collapses at five ( $26.3 \pm 20.2$ ); under the controller every cell stays alive and degrades gracefully ( $90.7 \rightarrow 88.9 \rightarrow 88.1$  sequential,  $95.3 \rightarrow 94.8 \rightarrow 89.6$  static), with the realised  $\lambda_\ell$  dropping to  $3 \times 10^{-6}$  in the starved middle layers. The residual toll is then architectural, and a biologically unexceptional bypass removes it. ResNet-style skip synapses  $S^\ell$  carry  $a^{\ell-2}$  into each deep hidden layer; they are learned by the same local product with a Kolen–Pollack feedback twin and the shared per-layer decay, and replay-masked under the same pre-or-post-asleep rule (Appendix A). They restore the thick error path, and the five-hidden-layer system reaches  $91.0 \pm 0.3$  sequential and  $95.8 \pm 0.3$  static (six seeds), against the two-layer default’s  $91.6/94.1$  and the two-layer controller’s  $90.6/93.5$ : 0.6 sequential points below the default and 1.7 above it on the static axis, with  $95.8 \pm 0.1$  at depth four. The bypass alone also rescues the fixed decay ( $90.4/93.5$  at depth five), which indicates that error-path attenuation, rather than decay mis-tuning as such, is a principal contributor to the depth collapse; the controller still adds its usual static margin on top of the bypass. We did not re-tune the fixed decay per depth: the claim is that the controller needs no re-tuning, not that no fixed value would serve. A BP+ER reference at the same widths is depth-insensitive ( $88.9 \rightarrow 87.9$  sequential) but forgets  $2\times$  more ( $F \approx 13\text{--}14$  vs.  $7\text{--}10$ ); the local system with skips and the controller beats it at every depth tested. The two starvations are not interchangeable, however. On split CIFAR-100 depth is a net loss that skips only half refund (development-phase numbers: four hidden layers  $1.9/1.6 \rightarrow 3.8/3.0$  with skips, against  $5.7/6.0$  at two), whereas *width* at the short chain length lifts the local system to  $5.8 \pm 0.9/7.7 \pm 1.1$  (1024–512). The thin-signal regime wants capacity, not depth, and BP+ER’s remaining lead there ( $10.1 \pm 0.4$ ) comes with twice the forgetting.

**What the controller must never see, and what it does not do.** Composition follows one rule: the drive estimate must be fed only *unamplified waking updates*. Inflated one-hot targets and the offline night’s  $3\times$ -gain batches both push the EMA up until  $\lambda_\ell$  hits its clip and erases day learning (in development, night  $\times$  controller collapsed to  $60.2 \pm 2.8$  against  $90.9 \pm 3.2$  for the night alone), whereas mechanisms that respect this rule compose cleanly (anchor and bout gate under the controller kept their usual signature in development). The controller also deliberately does not take over  $\lambda$ ’s second role. On small tabular problems the decay serves as a *capacity* regulariser against overfitting (cf. Appendix B), information that no drive ratio contains; where generalisation rather than alignment sets the correct decay, a tuned constant should keep the job.

## 8 Discussion

**Why micro-batches work.** A replay micro-batch is a high-variance gradient estimate. Applied to a static coalition it perturbs the very units encoding the wake stream, and with shared optimiser state the perturbation compounds (the rotation-free unmasked curve of Figure 4). Two things defuse it. Alternating the coalition, which is what the rotation does, is where most of the accuracy comes from: with it, even unmasked replay holds  $91.5$  at batch 16 and  $89.3$  at batch 4. Confining the update to the mask adds  $0.1\text{--}0.3$  points at those batch sizes, shows none of the severe failures that unmasked replay shows in half the seeds at batch 2 (Appendix H), and gives the guarantee that the present input’s

hidden computation, under its suppression mask, is untouched: exactly on the proven channels, and for all but  $\approx 0.3\%$  of waking samples per step in measurement (Propositions 1–2 and the Remark). The offline night has neither protection: with nothing clamping the network, a small noisy batch at  $3\times$  learning rate moves the whole weight state.

**Prerequisites beyond this substrate.** The construction is a recipe with two checkable prerequisites. If a system’s representation is sparse enough that a batch leaves most units silent, and its learning rule yields a local error signal, consolidation can run in the silent degrees of freedom between the steps of the stream, with no separate offline phase, under a conditional guarantee on the computation in use, and at a higher total compute ( $6\times$  the night’s wall-clock, unoptimised). Both prerequisites can be checked on a given architecture. Top- $k$  routed mixture-of-experts layers expose, per token, the experts a token leaves silent, but whether the routes of a whole batch leave enough experts idle is a measurement (load balancing pushes against it), and the local error signal is what such a system would still have to supply.

**Correspondence with biology.** We treat the biology as a motivating analogy, not as a claim of mechanistic equivalence. The refractory rule mirrors the causal finding of [Driessen et al. \(2026\)](#) that ON/OFF *alternation*, not tonic reduction, discharges local sleep pressure and rescues memory: our soft-rotation control (a tonic reduction) is reliably worse, and the rotation-free silent mask, with no imposed off-periods at all, is worse still. The dissociation of triggers, homeostatic for waking consolidation and novelty-timed for full nights, is a testable prediction for biology. The regime dependence of the rotation price (a cost on easy streams, a gain under chronic underfitting) may bear on why local sleep in vivo both impairs behaviour ([Vyazovskiy et al., 2011](#)) and serves useful functions when appropriately timed.

**Limitations.** Configurations were selected on the official test splits (the development set) and frozen; every main comparison re-trains them with a held-out tenth of the training data removed from the stream and evaluates on it (two splits for the headline rows, sharing 9% of their samples), and exploratory results not re-run under this protocol are labelled development-phase. Development-set numbers sit 1.1 points above the held-out ones on average and rank-correlate with them at Spearman  $\rho = 0.97$ , a robustness check rather than a proof that selection bias is absent (Appendix I). The protocol is five epochs per task; a single pass keeps the lead (Table 1, Appendix G). All results are at MLP scale on MNIST/CIFAR variants with three seeds (six on the headline rows). BP+ER is an external reference rather than a state-of-the-art claim, and stronger replay baselines such as DER++ are not included; it keeps a 0.5-point lead on feature-based CIFAR with both sides at their best, a lead the substrate decomposition attributes to dense activation rather than to backpropagation itself. Rotation keeps a 1–2-point static price at the 10% fraction that no gate we tested removes without a sequential cost (Section 6). The same-substrate offline night is highly variable across seeds ( $89.0 \pm 3.5$  on the reported split,  $91.5 \pm 0.5$  on the second, where it trails the full system by only 0.3), which we report but do not explain. Finally, the novelty claims rest on our own literature search (August 2026); the individual primitives (refractory competition, sparse or null-space isolation, replay, adaptive scheduling) all have ancestors, and the claimed contribution is their combination into concurrent, exactly isolated consolidation.

## 9 Future Work

(i) The bimodal instability of the same-substrate offline night at the 10% fraction ( $89.0 \pm 3.5$ ), and what makes offline consolidation fragile on wide substrates. (ii) The bout gate’s residual static price (94.2 vs. 96.0 rotation-free, development-phase 5% numbers): the OFF-side refractory, utility exemption and synaptic anchoring are all ruled out as routes past it (Section 6), leaving mastery-annealed rotation depth untried. (iii) The narrow-vs-wide substrate interaction of offline nights at tiny buffers. (iv) Convolutional stacks. Depth and residual-style routing are answered (Section 7: five hidden layers at the two-layer level with locally learned skips and the controller; on 100-class streams width, not depth, is what pays), but convolution and a learned front-end are not, and the static-axis wall of the 100-class regime (6–8% under every substrate change) indicates that the front-end is the binding constraint there. (v) A throughput theory of the isolated channel that predicts the required replay cadence from  $\rho_\ell$ , code overlap and the interference rate. (vi) The interaction between rotation and spike calibration, which was optimiser-dependent in development and small on the held-out split. (vii) A drive estimate that survives amplified update streams (inflated targets, night gains), which currently defeat the controller’s clip.

## 10 Conclusion

In the evaluated class-incremental MLP settings, consolidation does not require a separate offline phase. The invariance the isolation provides does not protect a finished forward pass; it closes the channel through which consolidation would perturb the units the present stream is using. Its accuracy value is 2.6 points without rotation and 0.1–0.3 with it (Table 1). Once the coalition alternates, the live units tolerate unmasked replay down to four-sample micro-batches; what

isolation adds is the guarantee that they need not, and, at two-sample batches where that tolerance runs out, graceful degradation where the unmasked control failed severely in half the seeds. With isolated replay, refractory rotation of the active coalition, homeostatic burst triggering and a relative-novelty gate on the rotation itself, a biologically constrained learner consolidates between the steps of its stream. It is competitive with the tested offline-night schedules on both held-out splits (above the best night by 2.5 on the first and by 0.3 on the second) and consistently above BP+ER and unmasked local replay, at  $2.2\times$  the night's replay samples (and, on the first split, at  $1.1\times$ ); it pays one to two points on i.i.d. streams, gains outright on harder data, and runs at a twentieth of dense-rate inference energy when spiking. The offline night retains one irreducible role, consolidating very small episodic stores, which sharpens rather than blurs the division of labour between wake and sleep.

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## A One training step, the three channels and the skip path

Algorithm 1 lists one waking step of the default system with the state each quantity is read from. Two conventions matter for the guarantees. First, the awake sets are read from the waking forward pass, i.e. from the weights before the unmasked waking update, while the masked replay step acts on the updated weights; Propositions 1–2 hold verbatim for a mask computed on the weights being updated, and the drift reported in Section 4.2 is measured under the implemented order. Second, the replay micro-batch is inferred with the suppression removed, so a refractory unit may fire on replay and receive the update (Corollary 1 makes that update invisible to the waking batch for any magnitude), and a burst of  $n_b$  micro-batches reuses the mask of its waking batch. The waking step is  $\eta_w = \eta m/256$  for a waking batch of  $m$  samples (the per-epoch drift of a batch-256 protocol), the replay step  $\eta_r = 3\eta$  per micro-batch, with  $\eta = 0.02$  throughout.

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**Algorithm 1** One waking step of the default system (rotation always on, one burst per batch).

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**Require:** weights  $W_t$ , velocities  $v_t$ , suppression  $s(t)$ , buffer  $\mathcal{M}$ , pressures  $S$

- 1:  $a \leftarrow \text{forward}(X_t; W_t, s(t))$ ,  $\varepsilon \leftarrow \text{sweep}(a, y_t)$  Eqs. (3)–(4)
- 2:  $(W, v) \leftarrow \text{unmasked heavy-ball step on } (W_t, v_t) \text{ with } G^\ell = \varepsilon^\ell (a^{\ell-1})^\top$ , rate  $\eta_w$  wake update
- 3:  $\mathcal{A}^\ell \leftarrow \{i : \exists b, a_{i,b}^\ell \neq 0\}$  for every hidden layer;  $S_i \leftarrow S_i + \bar{a}_i(X_t)$  from the pre-update  $a$
- 4:  $M^\ell \leftarrow \text{Eq. (6), unit mask } \mathbf{1}[i \notin \mathcal{A}^\ell]$ , readout row and bias unmasked
- 5:  $s_i^\ell(t+1) \leftarrow \mathbf{1}[i \in \mathcal{A}^\ell]$  refractory rotation
- 6: write  $(X_t, y_t)$  into  $\mathcal{M}$  (reservoir)
- 7: **for** each of the  $n_b$  replay micro-batches  $(X_r, y_r) \sim \mathcal{M}$  **do**
- 8:  $a_r \leftarrow \text{forward}(X_r; W, s=0)$ ,  $\varepsilon_r \leftarrow \text{sweep}(a_r, y_r)$  suppression removed
- 9:  $(W, v) \leftarrow \text{masked step, Eq. (7), with } G^\ell = \varepsilon_r^\ell (a_r^{\ell-1})^\top$ , rate  $\eta_r$ , masks  $M^\ell$
- 10: **end for**
- 11:  $S_i \leftarrow 0$  for every  $i \notin \mathcal{A}^\ell$  if a burst ran discharge
- 12: **return**  $W_{t+1} = W$ ,  $v_{t+1} = v$ ,  $s(t+1)$

---

**The three channels.** Table 3 lists the three ways a synapse enters the mask (6) and what each guarantees; the rotation of Section 4.3 exists to move synapses from the second row into the third.

| Channel                                   | Condition                                       | Invisible to $X$                   | Source  |
|-------------------------------------------|-------------------------------------------------|------------------------------------|---------|
| presynaptic silent                        | $j \notin \mathcal{A}^{\ell-1}(X)$              | for any magnitude                  | Prop. 1 |
| postsynaptic naturally asleep, active pre | $i \notin \mathcal{A}^\ell(X)$ , $s_i^\ell = 0$ | under the margin $\tau_{k,b}^\ell$ | Prop. 2 |
| postsynaptic suppressed                   | $s_i^\ell = 1$                                  | for any magnitude                  | Cor. 1  |

**Table 3: The three consolidation channels** opened by the synapse mask.

**A numerical example.** On one sample with  $k = 2$  and  $\sigma(z) = (3, 2, 0.5, 0)$ , units 3 and 4 are asleep and  $\tau_k = 2$ . Any replay update with  $\sigma(z_3 + \delta_3) < 2$  and  $\sigma(z_4 + \delta_4) < 2$  leaves the activity vector  $(3, 2, 0, 0)$  unchanged, because the two winners’ own inputs are not touched; if unit 4 is refractory ( $s_4 = 1$ ) its bound is void and  $\delta_4$  may be arbitrarily large; and every synapse leaving unit 4 into the next layer may move by any amount, since its presynaptic activity is zero.

**Skip synapses.** For  $\ell \geq 3$  each deep hidden layer receives a bypass,  $z^\ell = W^\ell a^{\ell-1} + S^\ell a^{\ell-2} + b^\ell$ , with  $S^\ell$  under the same Dale projection and a feedback twin  $B_S^\ell \in \mathbb{R}^{n_{\ell+2} \times n_\ell}$  that carries the bypass target’s burst back to its source,

$$\varepsilon^\ell \leftarrow \varepsilon^\ell + \sigma'(z^\ell) \odot \left( (B_S^\ell)^\top [\kappa \tanh(\varepsilon^{\ell+2}/\kappa) \odot \mathbf{1}(a^{\ell+2} > 0)] \right),$$

$S^\ell$  and  $B_S^{\ell-2}$  learned by the local product of (5) with the target layer’s decay  $\lambda_\ell$ , and replay-masked by the same rule two layers apart,  $M_{S,ij}^\ell = \mathbf{1}[j \notin \mathcal{A}^{\ell-2}(X) \vee i \notin \mathcal{A}^\ell(X)]$ , so that Propositions 1–2 extend unchanged.

## B The price of the substrate

Table 4 traces the path from a dense backpropagation MLP to the substrate used throughout, on i.i.d. MNIST where each constraint’s cost is cleanest. The upper block is cumulative; the lower block replaces one ingredient at its stage and is to be compared with the matching cumulative row.

**Table 4: Substrate ablation** on i.i.d. MNIST (256–128 hidden core, three seeds; development phase, official test set). “Syn. ops” is the fraction of the dense MLP’s synaptic operations in one forward pass. Upper block: constraints added cumulatively; lower block: one ingredient replaced at its stage.

| Substrate                                                                  | Accuracy                        | Syn. ops  |
|----------------------------------------------------------------------------|---------------------------------|-----------|
| dense backpropagation MLP (autograd reference)                             | $97.7 \pm 0.1$                  | 1.00      |
| local burst-error learner, unconstrained                                   | $97.8 \pm 0.1$                  | 0.85      |
| + bounded error, Dale’s law, sign-concordant feedback (no transport)       | $97.1 \pm 0.0$                  | 0.85      |
| + $k$ -WTA 10%                                                             | $97.0 \pm 0.2$                  | 0.80      |
| + relaxation $T=5$                                                         | $97.0 \pm 0.1$                  | 0.80      |
| + 30% distance-dependent connectivity                                      | $96.5 \pm 0.2$                  | 0.24      |
| + burst gate and burst baseline                                            | $96.3 \pm 0.2$                  | 0.24      |
| + single-phase burst sweep replacing relaxation ( <i>final substrate</i> ) | $96.5 \pm 0.1$                  | 0.24      |
| random wiring instead of distance-dependent, equal density                 | $95.4 \pm 0.0$                  | 0.24      |
| one settling step ( $T=1$ ) instead of five                                | $17.9 \pm 6.8$                  | 0.80      |
| weight transport restored (mirroring, no KP decay)                         | $96.3 \pm 0.2$                  | 0.24      |
| Kolen–Pollack decay removed                                                | $96.3 \pm 0.1$                  | 0.24      |
| heavy-ball SGD instead of Adam ( $\eta = 0.02$ )                           | $95.2 \pm 0.3$                  | 0.24      |
| $k$ -WTA 5% / 2% instead of 10%                                            | $95.9 \pm 0.1$ / $94.2 \pm 0.2$ | 0.79/0.78 |
| multiplicative burst coding                                                | $78.8 \pm 1.9$                  | 0.24      |
| training on $T_s=8$ spike counts                                           | $80.0 \pm 0.8$                  | 0.24      |

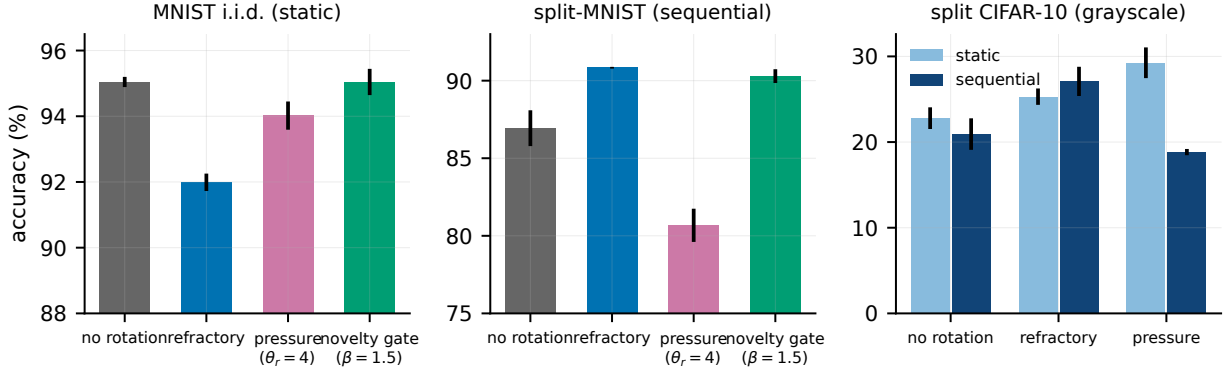
The cumulative ladder prices the entire biological constraint set at 1.2 points below the dense backpropagation reference ( $97.7 \rightarrow 96.5$ ) for  $4\times$  fewer synaptic operations per pass; the local burst-error objective itself costs nothing ( $97.8$  unconstrained). No single constraint is expensive: the bounded-error/Dale/no-transport trio costs 0.7, 10%  $k$ -WTA one tenth of a point, and the 70% connectivity cut half a point, of which distance-dependent wiring recovers +1.1 over random wiring at equal density. The lower block adds three cautions. Settling cannot merely be shortened ( $T=1$  collapses to 17.9); the single-phase burst sweep is the principled replacement, matching the  $T=5$  accuracy in a single pass. The Kolen–Pollack decay and even fully restored weight transport are accuracy-neutral at this scale; the decay is useful as a regulariser on harder inputs, not as an aligner. And the two spiking-flavoured training schemes fail outright: spiking is affordable at *inference* (the energy measurements above), not as a training-time code.

Finally, the two tables disagree about the optimiser in an instructive way. On the plain substrate SGD *loses* 1.3 points to Adam ( $95.2$  vs.  $96.5$ ), yet inside the isolated-consolidation loop the two tie (Table 1). Per-synapse adaptive state is an asset while every synapse trains on every batch, and no longer one once updates become intermittently mask-confined. The optimiser is thus not a neutral implementation detail: its advantage on the static substrate does not carry into the consolidating system, where the least state suffices.

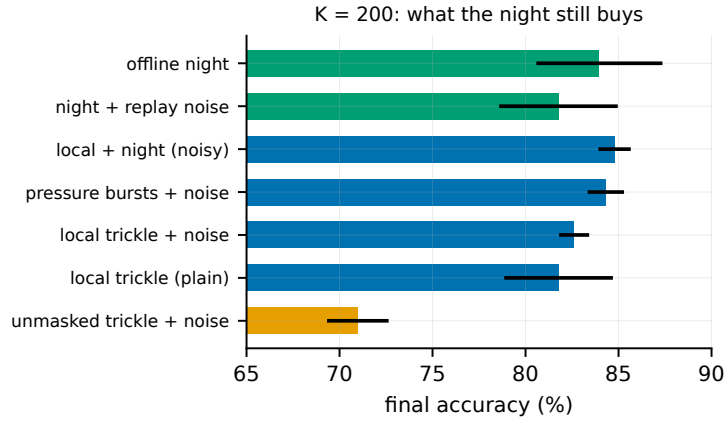
## C The per-batch rotation gates under masked Adam

Appendices C–G report development-phase numbers (official test set) from the earlier 5% phases, the small-buffer and down-selection studies and the learning-rate sweeps; they were not re-run under the held-out protocol and support mechanism orderings only, except where a paragraph is marked held-out (the single-pass streams of Appendix G).

Figure 7: always-on rotation costs 1.3–3.0 points on i.i.d. MNIST ( $95.0 \rightarrow 92.0$ – $93.7$ ). Gating rotation by *unit-level pressure* only trades the axes (static  $91.6 \rightarrow 94.0$  as sequential  $88.8 \rightarrow 80.7$ ; no dominant point). The *relative novelty* gate (8) removes the price at no sequential cost on MNIST: static  $95.0 \pm 0.4$  (rotation on 1.4–17% of batches), sequential  $90.3 \pm 0.4$  ( $\beta=1.5$ ) or  $90.9 \pm 0.6$  ( $\beta=1.1$ ). On split CIFAR-10 the price *reverses*: rotation gains +2.5 (refractory) to +6.5 points (pressure-selected rest) on the static axis, because forced rotation acts as a regulariser in the underfit regime, while the pure novelty gate (8) *misfires* there (sequential  $19.1 \pm 1.1$ : fast/slow  $\rightarrow 1$  on a stream that never becomes predictable, so rotation shuts off exactly where it helps). The gate with the unmastered clause (9) resolves this with one setting across all four regimes: split CIFAR-10 sequential  $27.2 \pm 0.7$  and static  $25.3 \pm 1.0$  (both = always-on rotation), split-MNIST  $90.4 \pm 0.9$ , i.i.d. MNIST  $94.3 \pm 0.2$  at  $\gamma=0.5$ . It retains a residual 0.7-point static cost relative to the pure novelty gate on easy streams, the price of the unmastered clause. All gate numbers in this appendix use masked Adam; under heavy-ball SGD the gate must be committed to minimum-duration bouts (Section 6).



**Figure 7: Rotation: price, trade-off, and gate.** Left: on i.i.d. MNIST the novelty gate restores no-rotation accuracy. Middle: on split-MNIST rotation is required and the gate keeps it. Right: on split CIFAR-10 (grayscale) rotation is a gain on *both* axes.



**Figure 8:  $K = 200$  mechanism triangulation.** Isolated rotating replay beats the unmasked trickle, diversity helps only during wake, and the irreducible advantage of the night at tiny buffers is its interference-free window.

## D Small buffers: what the night still buys

At  $K=200$  the night at first appeared to keep a  $\sim 2.6$ -point edge; most of it turns out to be a substrate mismatch (Figure 8). On the *same* 512–256/5% substrate, the night reaches  $84.0 \pm 3.4$  while noisy pressure-burst local sleep reaches  $84.3 \pm 1.0$  and local+night  $84.8 \pm 0.9$ : parity, with three-fold smaller variance for the wake-side scheme. The strongest tiny-buffer number remains the narrower 256–128/10% night (86.9), so the offline window retains an advantage only in combination with narrow substrates. Two mechanism results hold regardless. Replay diversity (noise  $\sigma=0.2$ ) is specifically a wake-side lever: it lifts every local scheme ( $81.8 \pm 2.9 \rightarrow 84.3 \pm 1.0$ ) and *hurts* the night ( $84.0 \rightarrow 81.8 \pm 3.2$ ). And *unmasked*, rotation-free waking replay collapses ( $71.0 \pm 1.7$ ), so isolated rotating replay beats it at every buffer size we tested.

## E Sleep-anchored down-selection

Motivated by the synaptic homeostasis hypothesis (Tononi and Cirelli, 2014), we anchored structural plasticity (prune the weakest fraction of live synapses, regrow the same number with a distance bias) to consolidation windows, using the Kolen–Pollack decay of (5) as the natural weakening force on unreplayed synapses. The outcome splits three ways at matched turnover ( $\approx 5\%$  per epoch). On the *continuous* local-sleep substrate pruning is free (sleep-anchored  $90.9 \pm 0.2$ , clocked  $90.6 \pm 0.4$ , none  $90.8 \pm 0.1$ ). On the pure *night* substrate, pruning immediately after each night helps substantially ( $87.2 \pm 3.8$  vs.  $84.1 \pm 1.3$ ): offline down-selection rescues the wide-sparse night. On the *episodic-burst* substrate pruning is harmful ( $84.6 \pm 0.5$  burst-anchored,  $86.6 \pm 1.3$  clocked, vs.  $89.9 \pm 0.1$  without): when replay is

rare, synapses are cut before enough consolidation events have protected them. Down-selection, like replay itself, needs either density or an offline window.

## F Learning-rate sensitivity

To ensure that the optimiser comparison is not an artefact of tuning SGD’s  $\eta$  while leaving Adam’s at its default, both were swept. At the 10% fraction (three seeds), heavy-ball SGD under always-on rotation holds  $92.2 \pm 0.7/93.3 \pm 0.2$  at  $\eta=0.01$ ,  $92.7 \pm 0.3/94.7 \pm 0.3$  at 0.02 (the joint optimum) and  $91.5 \pm 0.6/94.7 \pm 0.3$  at 0.05, decaying at 0.005 ( $91.3/91.3$ ) and destabilising at 0.1 ( $79.5 \pm 17.3$ ), while masked Adam has its optimum at the default ( $91.8 \pm 0.6/94.6 \pm 0.2$  at  $10^{-3}$ ;  $89.9/89.9$  at  $3 \cdot 10^{-4}$ ,  $90.9/94.3$  at  $3 \cdot 10^{-3}$ ). The comparison of Table 1 is therefore at each optimiser’s best rate. The sweep at the 5% fraction tells the same story. There, under always-on rotation, SGD holds a sequential plateau over  $\eta \in [0.01, 0.02]$  ( $91.9 \pm 0.5/92.0 \pm 0.6$ ) and a static plateau over  $[0.02, 0.05]$  ( $93.3\text{--}93.7$ ), decaying gracefully outside ( $86.3/89.2$  at  $\eta=0.1$ ), with  $\eta=0.02$  the joint optimum. Masked Adam, swept over a tenfold range, has its sequential optimum exactly at the default ( $90.8 \pm 0.1$  at  $10^{-3}$ ;  $88.6$  at  $3 \cdot 10^{-4}$ ,  $88.9$  at  $3 \cdot 10^{-3}$ ), and no Adam setting reaches the heavy-ball joint point ( $3 \cdot 10^{-3}$  buys  $93.6$  static at a three-point sequential cost). Unmasked SGD is poor and unstable at *every* rate ( $84\text{--}88$ , seed deviations  $1.8\text{--}4.1$ , three- to eight-fold those of the masked runs), so the stabilising role of isolation plus rotation (the unmasked runs are rotation-free) holds across  $\eta$ . The bout-committed gate inherits the plateau, and the optimum shifts one notch down on CIFAR ( $\eta=0.01$ ).

## G Probes of the residual static price

**Single-pass streams.** With one epoch per task instead of five (everything else as in the headline cells, three seeds), refractory local sleep reaches  $91.8 \pm 0.2$  ( $F=5.7$ ) against  $91.6$  with five epochs, BP+ER  $88.7 \pm 0.3$ , the offline night  $75.1 \pm 3.5$  (one night per task is too few), unmasked interleaved replay  $76.0 \pm 20.1$  and no buffer  $18.1$  (held-out split): the proposed system keeps its lead over every tested control in a single pass, while the offline night falls below BP+ER.

**Bout length and the OFF-side refractory.** Bout length is a genuine control parameter:  $M=512$  (duties  $0.61/0.38$ ) gives  $91.7 \pm 0.4/94.0 \pm 0.5$ , already recovering most of what the per-batch gate loses, while  $M=4096$  drives the static duty to  $0.94$  and the refund vanishes ( $93.4 \pm 0.1$ ). The static ceiling of silent isolation ( $96.0$ ) stays out of reach, because rotation keeps a residual price even in bouts; the committed gate buys regime-adaptivity, not a static gain. Completing the flip-flop with an OFF-side refractory (once a bout expires, triggers are ignored for  $M_{\text{off}}$  batches) is a clean null result. Short refractories ( $M_{\text{off}} \leq 1024$ ) never engage on the static stream, since a bout expires only after its trigger has been quiet for  $M$  batches and static re-triggering is therefore not flicker to begin with; a long one ( $M_{\text{off}}=4096$ ) lowers the static duty to  $0.53$  without moving static accuracy ( $94.0 \pm 0.7$ ) and monotonically erodes the sequential axis ( $92.1 \rightarrow 91.0$ , occasionally masking a task switch). Together with the  $M=512$  cell (duty  $0.38$ , static  $94.0$ ), this locates the residual static price in the rotation of settled coalitions itself, not in how often the gate fires.

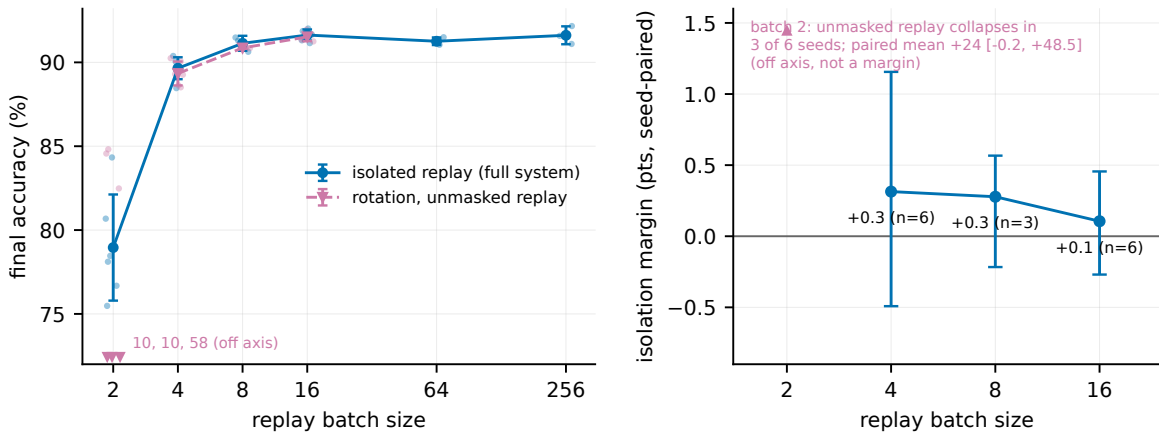
**Structural exemption and synaptic anchoring.** *Utility-exempt rotation* spares the top- $q$  units by long-term use trace from ever being benched, so the settled coalitions stay awake. It traces a smooth trade-off between always-on rotation and the rotation-free system ( $q=0.10$ :  $90.9 \pm 0.1/94.2 \pm 0.2$ ;  $q=0.25$ :  $89.1 \pm 0.4/95.4 \pm 0.2$ ), but the bout gate beats the  $q=0.10$  point by  $+1.1$  sequential at equal static. *Temporal commitment dominates structural exemption*, which repeats at the utility level what unit-level pressure gating showed earlier (Figure 7): any structural narrowing of the rotation pool is paid for in replay-channel width.

*Two-timescale synapses* give every weight a slow anchor in the spirit of synaptic consolidation cascades (Benna and Fusi, 2016): the fast weight decays toward its anchor at rate  $\lambda$  while the anchor absorbs it at rate  $\mu$ , ticking only on waking updates so the exactness of the replay path is untouched. The strong hypothesis, that settled function living in the anchor would dissolve rotation’s static price, is falsified: across  $\lambda \in [3 \cdot 10^{-4}, 3 \cdot 10^{-3}]$ ,  $\mu \in [10^{-4}, 10^{-3}]$  no cell exceeds the anchor-free static accuracy, and strong coupling hurts both axes (the anchor becomes a lagging drag). Weak symmetric coupling ( $\lambda=\mu=3 \cdot 10^{-4}$ ), however, acts as a *sequential stabiliser*:  $92.4 \pm 0.2$ , the best sequential cell of the 5% grid, with the seed variance halved, at static parity, weakly dominating the 5% always-on point. Anchor and bout gate compose sub-additively ( $92.3 \pm 0.3/93.8 \pm 0.5$ , between the parents).

## H Isolation across replay batch sizes

Figure 9 extends the left panel of Figure 4 to two-sample replay micro-batches and shows, per batch size, the seed-paired margin of the full system over rotation with unmasked replay (held-out split; six seeds at batches 2, 4 and 16, three at

8). From 16 down to 4 the margin is 0.1  $[-0.3, +0.5]$ , 0.3  $[-0.2, +0.6]$  and 0.3  $[-0.5, +1.2]$ : isolation costs nothing and buys nothing measurable in mean accuracy while the rotation keeps the live coalition tolerant of unmasked replay. At batch 2 the replay gradient is too noisy for either system. The isolated one falls to  $79.0 \pm 3.2$  (forgetting  $21.5 \pm 4.2$  against 6.4 at batch 16), with every seed between 75.5 and 84.3; the unmasked one bifurcates, with three seeds finishing at 82.5–84.8, slightly *above* their isolated counterparts (+0.3, +1.8, +8.1), one at 58.2 and two at chance (9.9). Leaking replay onto the live coalition is therefore not a pure loss, since the surviving seeds show that the extra plasticity can help, but it exposes a severe-failure mode (final accuracy below 60%: two seeds at chance, one at 58.2) that the isolated system did not show in six seeds. With the current input’s hidden computation held invariant, the noisiest replay degraded the system gracefully in every seed tested. Six seeds without a severe failure bound its frequency but do not prove the mode absent, and the masked system itself destabilises at replay gain 10 (Section 6). The paired mean at batch 2 ( $+24 [-0.2, +48.5]$ ) describes the unmasked failures, not a margin, and is kept out of the ranges quoted in the main text.



**Figure 9: Isolation across replay batch sizes (held-out split).** **Left:** split-MNIST accuracy against replay batch size for isolated replay (the full system) and for rotation with unmasked replay, every seed shown; at batch 2 no unmasked mean is drawn because three of six seeds fail severely (values at the axis floor). **Right:** the seed-paired margin of isolation with a 95% bootstrap interval at the batch sizes where the unmasked run does not collapse.

## I Development-phase numbers and a second held-out split

Configurations were selected on the official test splits (the development set) and frozen; the comparisons in the main text re-train them on nine tenths of the training data and evaluate on the held-out tenth, which took part in no selection. Table 5 places the development-set numbers beside the held-out ones and beside a second held-out tenth for the headline rows (drawn with a different seed; the two tenths share 9% of their samples, so they are not disjoint test sets). Over the 87 sequential configurations run under both protocols, reported numbers sit 1.1 points below the development numbers on average (a smaller training set and a distribution-matched evaluation set both contribute) and rank-correlate with them at Spearman  $\rho = 0.97$  (a robustness check, not a proof that selection bias is absent); every ordering carried by the paper holds on both held-out splits, with one magnitude that does not: the same-substrate night, bimodal on the first split ( $89.0 \pm 3.5$ ), reaches  $91.5 \pm 0.5$  on the second and trails the full system by 0.3 there. The adaptive decay’s static margin on MNIST is likewise split-dependent ( $93.5 \pm 4.3$  with one collapsed seed on the first split,  $95.2 \pm 0.4$  on the second). Table 6 gives the per-task accuracy matrix of the headline configuration on the reported split.

**Table 5: Development-set (official test split) numbers beside the held-out numbers and a second held-out split.** Sequential accuracy, with the static axis on its own row where the paper reports it; six seeds on the reported split for the headline rows, three elsewhere. Lower block: raw split CIFAR-10.

|                                                 | development set | held-out split 1 | held-out split 2 |
|-------------------------------------------------|-----------------|------------------|------------------|
| always-on rotation (default)                    | $92.7 \pm 0.3$  | $91.6 \pm 0.3$   | $91.8 \pm 0.4$   |
| static                                          | $94.7 \pm 0.3$  | $94.1 \pm 0.2$   | $94.1 \pm 0.6$   |
| bout-committed gate                             | $92.3 \pm 0.2$  | $91.8 \pm 0.3$   | $92.0 \pm 0.6$   |
| rotation alone (unmasked replay)                | $92.1 \pm 0.5$  | $91.5 \pm 0.3$   | $91.8 \pm 0.6$   |
| static                                          | $94.7 \pm 0.1$  | $93.7 \pm 0.3$   | $93.7 \pm 0.3$   |
| silent mask (no rotation)                       | $88.8 \pm 0.8$  | $88.0 \pm 0.7$   | $88.6 \pm 1.3$   |
| unmasked interleaved ER (SGD)                   | $87.0 \pm 1.1$  | $85.4 \pm 3.5$   | $80.0 \pm 14.1$  |
| BP + ER                                         | $89.3 \pm 0.8$  | $88.8 \pm 0.3$   | $88.5 \pm 0.7$   |
| offline night, same substrate                   | $90.9 \pm 3.2$  | $89.0 \pm 3.5$   | $91.5 \pm 0.5$   |
| offline night, narrow substrate                 | $90.7 \pm 0.4$  | $89.2 \pm 1.7$   | $89.3 \pm 0.6$   |
| adaptive $\lambda$ (drive)                      | $91.7 \pm 0.2$  | $90.6 \pm 0.6$   | $90.8 \pm 0.1$   |
| static                                          | $96.1 \pm 0.2$  | $93.5 \pm 4.3$   | $95.2 \pm 0.4$   |
| five hidden layers + skips + adaptive $\lambda$ | $91.5 \pm 0.4$  | $91.0 \pm 0.3$   | $91.1 \pm 0.3$   |
| static                                          | $96.2 \pm 0.3$  | $95.8 \pm 0.3$   | $95.7 \pm 0.1$   |
| no buffer                                       | $19.6 \pm 0.0$  | $19.4 \pm 0.0$   | $19.5 \pm 0.0$   |
| <i>split CIFAR-10 (raw)</i>                     |                 |                  |                  |
| rotation                                        | $29.5 \pm 0.8$  | $28.9 \pm 1.0$   | $29.7 \pm 0.9$   |
| offline night                                   | $26.2 \pm 0.6$  | $25.7 \pm 1.2$   | $26.5 \pm 0.4$   |
| BP + ER                                         | $25.1 \pm 1.5$  | $24.8 \pm 0.3$   | $25.8 \pm 0.8$   |
| unmasked ER                                     | $17.0 \pm 6.1$  | $15.4 \pm 4.2$   | $15.5 \pm 4.5$   |
| no buffer                                       | $16.4 \pm 0.2$  | $16.3 \pm 0.1$   | $16.6 \pm 0.0$   |

**Table 6: Per-task accuracy matrix** of the headline configuration on split-MNIST (held-out split, mean over six seeds): row = after training task  $i$ , column = accuracy on task  $j$ . Forgetting  $F$ , the mean of the diagonal minus the last row over tasks 1–4, is  $6.4 \pm 0.6$ ; the drop is concentrated on tasks 2–3 and the first task is retained.

| after task | T1   | T2   | T3   | T4   | T5   |
|------------|------|------|------|------|------|
| 1          | 99.8 |      |      |      |      |
| 2          | 98.7 | 97.2 |      |      |      |
| 3          | 97.7 | 92.7 | 97.0 |      |      |
| 4          | 97.5 | 89.6 | 92.9 | 97.5 |      |
| 5          | 96.7 | 88.1 | 89.7 | 91.5 | 91.8 |